Games in Finite, Well-Mixed Populations

For the Payoff Matrix $\mathbf{M} = \begin{array}{cc} & \begin{array}{cc} C & D \end{array} \\ \begin{array}{c} C \\ D \end{array} & \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{array}$

$b<c$ and $a>d$

No. of C players = $\mathbf{i}$

No. of D players = $\mathbf{N-i}$

Prob. that C interacts with another C = $(i-1)/(N-1)$

Prob. that C interacts with another D = $(N-i)/(N-1)$

Prob. that D interacts with another C = $i/(N-1)$

Prob. that D interacts with another D = $(N-i-1)/(N-1)$

Expected payoff to C when it interacts with C = $a(i-1)/(N-1)$

Expected payoff to C when it interacts with D = $b(N-i)/(N-1)$

Total expected payoff for C : $\mathbf{F_i = (a(i-1)+b(N-i))/(N-1)}$

Total expected payoff for D : $\mathbf{G_i = (ci+d(N-i-1))/(N-1)}$

Define fitness of **C** as : $f_i = 1- w + w F_i$; fitness of **D** as : $g_i = 1- w + w G_i$

$w=1 \rightarrow$ strong selection; fitness completely determined by interactions

$w=0 \rightarrow$ no selection between C & D

$w<<1 \rightarrow$ weak selection

Revisiting the ESS condition for large populations

When can a population of D players avoid being invaded by a single mutant C-type?

Selection opposes C invading D: Fitness of a single C-type < Fitness of (N-1) D-types

$f_1 < g_1$ ➔ $\boxed{b(N-1) < c + d(N-2)}$ ➔ For N>>1: **b < d** **;** For N=2: **b<c** ⟺

Infinite population result
D is an ESS iff:
b<d **or** if b=d, a<c

When can a population of C players avoid being invaded by a single mutant D-type?

Selection opposes D invading C: Fitness of a single D-type **<** Fitness of (N-1) C-types

c

$g_{N-1} < f_{N-1}$ ➔ $\boxed{(N-1) < b + a(N-2)}$ ➔ For N>>1: **a > c;** For N=2: **c<b** ⟺

Infinite population result
C is an ESS iff:
a>c **or** if a=c, b>d

Essential to consider fixation probability in finite populations to determine the ESS

$\rho_C > \frac{1}{N} \Rightarrow$ Selection favours C replacing D

$\rho_C < \frac{1}{N} \Rightarrow$ Selection opposes C replacing D

$$f_1 = 1 - w + wF_1$$
$$g_1 = 1 - w + wG_1$$

$$F_1 = \frac{a(1-1) + b(N-1)}{(N-1)} = b$$

$$G_1 = \frac{c \cdot 1 + d(N-1-1)}{N-1} = \frac{c + d(N-2)}{(N-1)}$$

$$f_1 < g_1$$
$$F_1 < G_1$$
$$b < \frac{c + d(N-2)}{(N-1)} \Rightarrow \boxed{(N-1)b < c + d(N-2)}$$

$$f_{N-1} = 1 - w + wF_{N-1}$$
$$g_{N-1} = 1 - w + wG_{N-1}$$

$$F_{N-1} = \frac{a(N-1-1) + b(N-N+1)}{N-1} = \frac{a(N-2) + b}{N-1}$$

$$g_{N-1} < f_{N-1}$$

$$G_{N-1} = \frac{c(N-1) + d(N-N+1-1)}{N-1} = c$$

$$= c < \frac{a(N-2) + b}{(N-1)}$$

$$= \boxed{c(N-1) < a(N-2) + b}$$

Moran Process in Games in Finite Populations

C's and D's are picked for reproduction with a probability proportional to their mean fitness and for death randomly.

Probability of picking **C** for reproduction and **D** for death : $p_{i,i+1} \equiv \alpha_i = \left(\frac{i f_i}{i f_i + (N-i) g_i}\right)\left(\frac{N-i}{N}\right)$

Probability of picking **D** for reproduction and **C** for death : $p_{i,i-1} \equiv \beta_i = \left(\frac{(N-i) g_i}{i f_i + (N-i) g_i}\right)\left(\frac{i}{N}\right)$

$$\gamma_i = \frac{\beta_i}{\alpha_i} = \frac{g_i}{f_i}$$

$$\rho_C = \frac{1}{1 + \sum_{k=1}^{N-1} \prod_{i=1}^{k} \frac{g_i}{f_i}} = x_1 \qquad \rho_D = \frac{\prod_{i=1}^{N-1} \frac{g_i}{f_i}}{1 + \sum_{k=1}^{N-1} \prod_{i=1}^{k} \frac{g_i}{f_i}} = 1 - x_{N-1} \qquad \frac{\rho_D}{\rho_C} = \prod_{i=1}^{N-1} \frac{g_i}{f_i}$$

In the limit w→0, $\rho_C > \frac{1}{N}$ leads to the inequality

$\boxed{a(N-2) + b(2N-1) > c(N+1) + d(2N-4)}$ which in the limit N>>1,reduces to $\boxed{a + 2b > c + 2d}$

For fixed, a,b,c,d, the above inequality gives a lower bound on the population size N

$\boxed{N > N_c}$

$$N_C = \frac{2a + b + c - 4d}{a + 2b - c - 2d}$$

Nc is the *minimum* size of the population necessary for selection to favour fixation of cooperators

$$\gamma_i = \frac{\beta_i}{\alpha_i} = \frac{g_i}{f_i} = \frac{1-\omega+\omega G_i}{1-\omega+\omega F_i} \rightarrow \lim_{\omega<<1} \gamma_i = (1-\omega+\omega G_i)(1+\omega-\omega F_i)$$

$$= 1-\omega(F_i-G_i) + O(\omega^2)$$

$$\frac{\rho_D}{\rho_C} = \prod_{i=1}^{k} \gamma_i = [1-\omega(F_1-G_1)][1-\omega(F_2-G_2)]\ldots = 1-\omega\sum_{i=1}^{k}(F_i-G_i) + O(\omega^2)$$

$$F_i - G_i = \frac{a(i-1)+b(N-i)}{(N-1)} - \frac{ci+d(N-i-1)}{(N-1)} = \frac{i(a-b-c+d)}{(N-1)} + \frac{(-a+bN-dN+d)}{(N-1)}$$

$$= iu + v$$

$$\sum_{i=1}^{k}(F_i-G_i) = u\sum_{i=1}^{k} i + vk = \frac{uk(k+1)}{2} + vk = \frac{uk^2}{2} + \left(\frac{u}{2}+v\right)k$$

$$\rho_C = x_1 = \frac{1}{1+\sum_{1}^{N-1}\prod_{i=1}^{k}\gamma_i} = \frac{1}{1+\sum_{k=1}^{N-1}\left(1-\omega\left[\frac{uk^2}{2}+\left(\frac{u}{2}+v\right)k\right]\right)}$$

$$\rho_C = \frac{1}{\cancel{1+(N-1)} - \omega\frac{u}{2}\frac{(N-1)N(2(N-1)+1)}{6} - \omega\left(\frac{u}{2}+v\right)\frac{N(N-1)}{2}}$$

$$= \frac{1}{N\left[1-\omega\left[\frac{u(N-1)(2N-1)}{12} + \frac{(u+2v)(N-1)}{4}\right]\right]}$$

Now, $u = \frac{(a-b-c+d)}{N-1}$; $v = \frac{-a+bN-dN+d}{(N-1)}$; $(u+2v) = \frac{a-b-c+d-2a+2bN-2dN+2d}{(N-1)}$

$$\therefore \frac{u(N-1)(2N-1)}{12} = \frac{(a-b-c+d)(2N-1)}{12} ;$$

$$= \frac{-a+3d-b(1-2N)-2dN-c}{(N-1)}$$

So, $\frac{(u+2v)(N-1)}{4} = \frac{-a+3d-b(1-2N)-2dN-c}{4}$

$$\rho_c = \frac{1}{N\left(1-\frac{\omega}{4}\left(\frac{(a-b-c+d)(2N-1)}{3} + (-c-a+3d-b(1-2N)-2dN)\right]\right.}$$

$$= \frac{1}{N}\left(1+\frac{\omega}{4}\Gamma\right) \quad \text{(Taylor approximation)}$$

$$= \frac{1}{N} + \frac{\omega}{4N}\Gamma$$

So, $\rho_c > \frac{1}{N}$ when $\boxed{\Gamma > 0}$

Note that, for Neutral Evolution, $\rho = \frac{1}{N}$ So, for fixation of C the condition $\rightarrow$ $\rho_c > \frac{1}{N}$

Now for $N > 0$

$$\Gamma = (a-b-c+d)\frac{2N}{3} - (2Nb - 2dN)$$ [Ignoring non multiple of N]

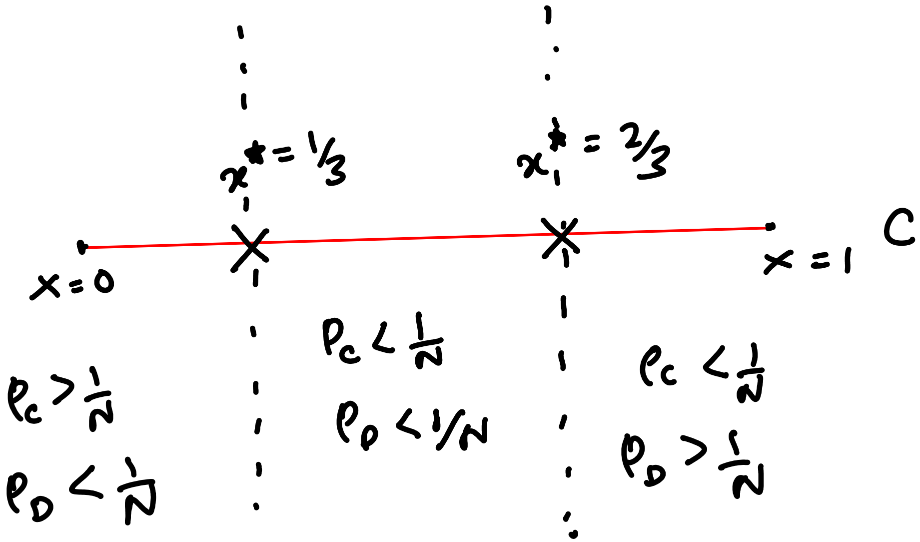
$\Gamma > 0 \Rightarrow (a-b-c+d)\frac{\cancel{2N}}{3} > \cancel{2N}(b-d)$

So, $a-b-c+d > 3b-3d \Rightarrow (a-c)+(d-b) > 3(b-d)$

$\Rightarrow (a-c)+(d-b) < 3(d-b)$

Remember, from replicator equation, mixed state equilibrium is $x^* = \dfrac{d-b}{(a-c)+(d-b)}$

So for $\Gamma > 0 \Rightarrow x^* < \frac{1}{3}$ for $\rho_c > \frac{1}{N}$



Finite N limit

$$\frac{(a-b-c+d)(2N-1)}{3} + (-c-a+3d-b(1-2N)-2dN) > 0$$

$$\Rightarrow (a-b-c+d)(2N-1) - 3c - 3a + 9d - 3b(1-2N) - 6dN > 0$$

$$\Rightarrow a(2N-1-3) + b(-2N+1-3+6N) > c(2N-1+3) + d(-2N+1-9+6N)$$

$$\Rightarrow 2a(N-2) + b(4N-2) > c(2N+2) + d(4N-8)$$

$$\Rightarrow \boxed{a(N-2) + b(2N-1) > c(N+1) + d(2N-4)}$$

$$\rho_c > \frac{1}{N} \Rightarrow \Gamma > 0 \Rightarrow a(N-2) + b(2N-1) > c(N+1) + d(2N-4)$$

So, $N(a+2b-c-2d) > 2a+b+c-4d$

$$\Rightarrow N > \underbrace{\frac{(2a+b+c-4d)}{(a+2b-c-2d)}}_{N_c} \Rightarrow \boxed{N > N_c}$$

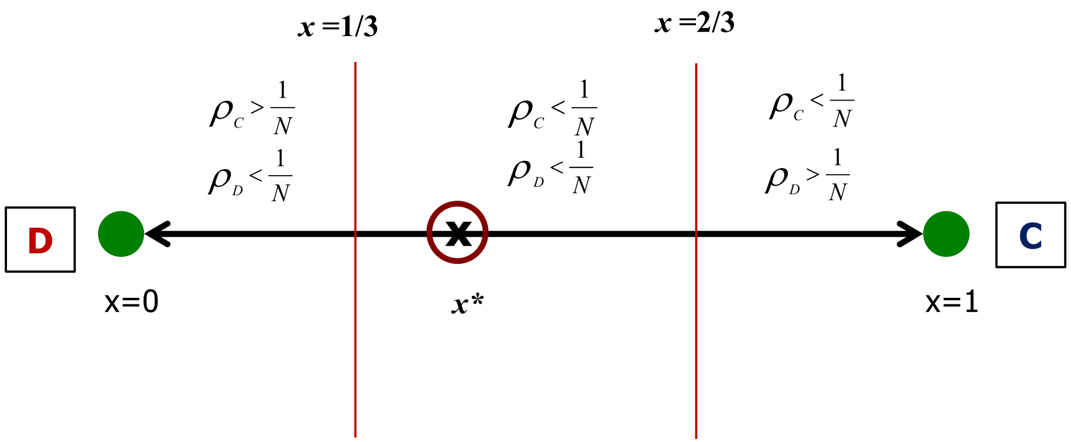
⋆ $C \rightarrow$ ESSN $\rightarrow \rho_c > \frac{1}{N}$ & $\rho_D < \frac{1}{N}$ || $D \rightarrow$ ESSN $\Rightarrow \rho_c < \frac{1}{N}$ & $\rho_D > \frac{1}{N}$

1/3 Law: Condition for the invasion probability of C > *neutral* invasion probability

$$a - c > 2(d - b)$$

$$x^* = \frac{d-b}{a-c+d-b} \quad \Rightarrow x^* < \frac{1}{3}$$

Condition on the mixed state equilibrium frequency obtained from replicator dynamics



Evolutionary Stability in Finite Populations

$$\begin{array}{c c} & \begin{array}{cc} C & D \end{array} \\ \begin{array}{c} C \\ D \end{array} & \begin{pmatrix} a & b \\ c & d \end{pmatrix} \end{array}$$

If a>c and b>d, **C** is a strict Nash equilibrium as well as an ESS and selection will always favour fixation of **C** and oppose fixation of **D** in a large population.

If a>c and b<d, both **C** and **D** is an ESS. According to the *infinite population* analysis, a small fraction of **C** mutants cannot invade a population consisting predominantly of **D** players.

What happens for finite populations ?

Condition for a strategy to be an ESS has to be modified for *finite* populations.

In a *finite* population of size N, a strategy **C** is an ***ESSN*** if → for finite population

(i) A single mutant of any other strategy has lower fitness than **C**

(ii) The fixation probability of *every other strategy* must be smaller than the fixation probability of a neutral strategy and the fixation probability of **C** must be larger than the fixation probability of a neutral strategy

C is an ***ESSN*** if $G_{N-1} < F_{N-1}$ and $\rho_C > \frac{1}{N}$ and $\rho_D < \frac{1}{N}$

D is an ***ESSN*** if $F_1 < G_1$ and $\rho_D > \frac{1}{N}$ and $\rho_C < \frac{1}{N}$

Evolutionary Stability in Finite Populations: Examples

For D to be an ESS:

Selection opposes C invading D: Fitness of a single C-type < Fitness of (N-1) D-types

$f_1 < g_1 \rightarrow$ $\boxed{b(N-1) < c + d(N-2)}$

Selection opposes C replacing D: $\rho_C < \frac{1}{N}$

$\boxed{a(N-2) + b(2N-1) < c(N+1) + d(2N-4)}$ for w<<1

For N>>1: **b < d** and **a+b<c+d**

For N=2: $f_1 < g_1$ ➔ b<c and $\rho_C < \frac{1}{2}$ ➔ b<c

$$M = \begin{array}{c c} & \begin{array}{cc} C & D \end{array} \\ \begin{array}{c} C \\ D \end{array} & \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{array}$$

Examples

$$M = \begin{array}{c c} & \begin{array}{cc} C & D \end{array} \\ \begin{array}{c} C \\ D \end{array} & \begin{bmatrix} 20 & 0 \\ 17 & 1 \end{bmatrix} \end{array}$$

Infinite population inference
Both C & D are an ESS

$f_1 < g_1$ ➔ -15<N

$\rho_C < \frac{1}{N}$ ➔ N<53

In finite populations
D is an ESS only for N<53

$$M = \begin{array}{c c} & \begin{array}{cc} C & D \end{array} \\ \begin{array}{c} C \\ D \end{array} & \begin{bmatrix} 1 & 28 \\ 2 & 30 \end{bmatrix} \end{array}$$

Infinite population inference
Only D is an ESS

$f_1 < g_1$ ➔ N>15

$\rho_C < \frac{1}{N}$ ➔ N>17

In finite populations
D is an ESS only for N>17

Whether a strategy is an ESS or not *depends on the population size*

Risk Dominance in Evolutionary Games

Risk Dominance: If both **C** and **D** is a strict Nash Equilibrium in the conventional sense
i.e. if a>c and d>b then which strategy has a higher fixation probability ?

$$\frac{\rho_D}{\rho_C} = \prod_{i=1}^{N-1} \gamma_i = \prod_{i=1}^{N-1} \frac{g_i}{f_i}$$

Approach is not correct

$$w<<1: \quad \gamma_i \simeq 1 - w(F_i - G_i) + o(w^2)$$

$$\frac{\rho_D}{\rho_C} = \prod_{i=1}^{N-1} (1 - w(F_i - G_i)) \simeq 1 - w\sum_{i=1}^{N-1}(F_i - G_i)$$

$$\sum_{i=1}^{N-1}(F_i - G_i) = \sum_{i=1}^{N-1} ui + v \qquad u = \frac{(a-b-c+d)}{N-1}; \qquad v = \frac{-a+bN-dN+d}{N-1}$$

$$\frac{\rho_D}{\rho_C} = 1 - \frac{w}{2}X; \quad \boxed{X = (a+b-c-d)N - 2a + 2d} \quad \boxed{X > 0 \Rightarrow \rho_C > \rho_D}$$

$$N >> 1: X > 0 \Rightarrow \boxed{a+b > c+d} \qquad x^* = \frac{d-b}{a-c+d-b} \quad \Rightarrow x^* < \frac{1}{2}$$

$$M = \begin{array}{c|cc} & C & D \\ \hline C & a & b \\ D & c & d \end{array}$$

Risk Dominance: If both **C** and **D** is a strict Nash Equilibrium in the conventional sense
i.e. if a>c and d>b then which strategy has a higher fixation probability ?

$$\frac{\rho_A}{\rho_B} = \prod_{i=1}^{N-1} \frac{f_i}{g_i} = \prod_{i=1}^{N-1} \frac{1-\omega+\omega f_i}{1-\omega+\omega G_i} \approx \prod_{i=1}^{N-1} (1-\omega+\omega F_i)(1+\omega-\omega G_i)$$

$$= 1 - \omega \sum_{i=1}^{N-1} (G_i - F_i) + O(\omega^2)$$

$$G_i - F_i = \frac{ci + d(N-i-1)}{N-1} - \frac{a(i-1)+b(N-i)}{(N-1)}$$

$$= \frac{i(c-d-a+b)}{N-1} + \frac{d(N-1)+a-bN}{(N-1)}$$

$$= i\bar{u} + \bar{v}$$

So,

$$\rightarrow \bar{u}\omega \sum_{i=1}^{N-1} i + \bar{v}\omega(N-1)$$

So, $$\frac{\rho_A}{\rho_B} = 1 - \bar{u}\omega \frac{N(N-1)}{2} - \bar{v}\omega(N-1)$$

$$= 1 + \frac{\omega}{2}\left[-\bar{u}N(N-1) - 2\bar{v}(N-1)\right]$$

$$\left.\begin{aligned} -\bar{u}(N-1)N &= (-c+d+a-b)N \\ -\bar{v}(N-1) &= -d(N-1)-a+bN \end{aligned}\right\}$$

$$\frac{\rho_A}{\rho_B} = 1 + \frac{\omega}{2}\left[(a-b-c+d)N - 2a - 2dN + 2d + 2bN\right]$$

$$= 1 + \frac{\omega}{2}\left[\underbrace{(a+b-c-d)N + 2(d-a)}_{>0}\right]$$

for $\frac{\rho_A}{\rho_B} > 0 \Rightarrow$

So,

$$N(a+b-c-d) > 2(a-d)$$

$$N(a-d) - 2(a-d) > N(c-b)$$

$$\Rightarrow \boxed{(N-2)(a-d) > N(c-b)}$$

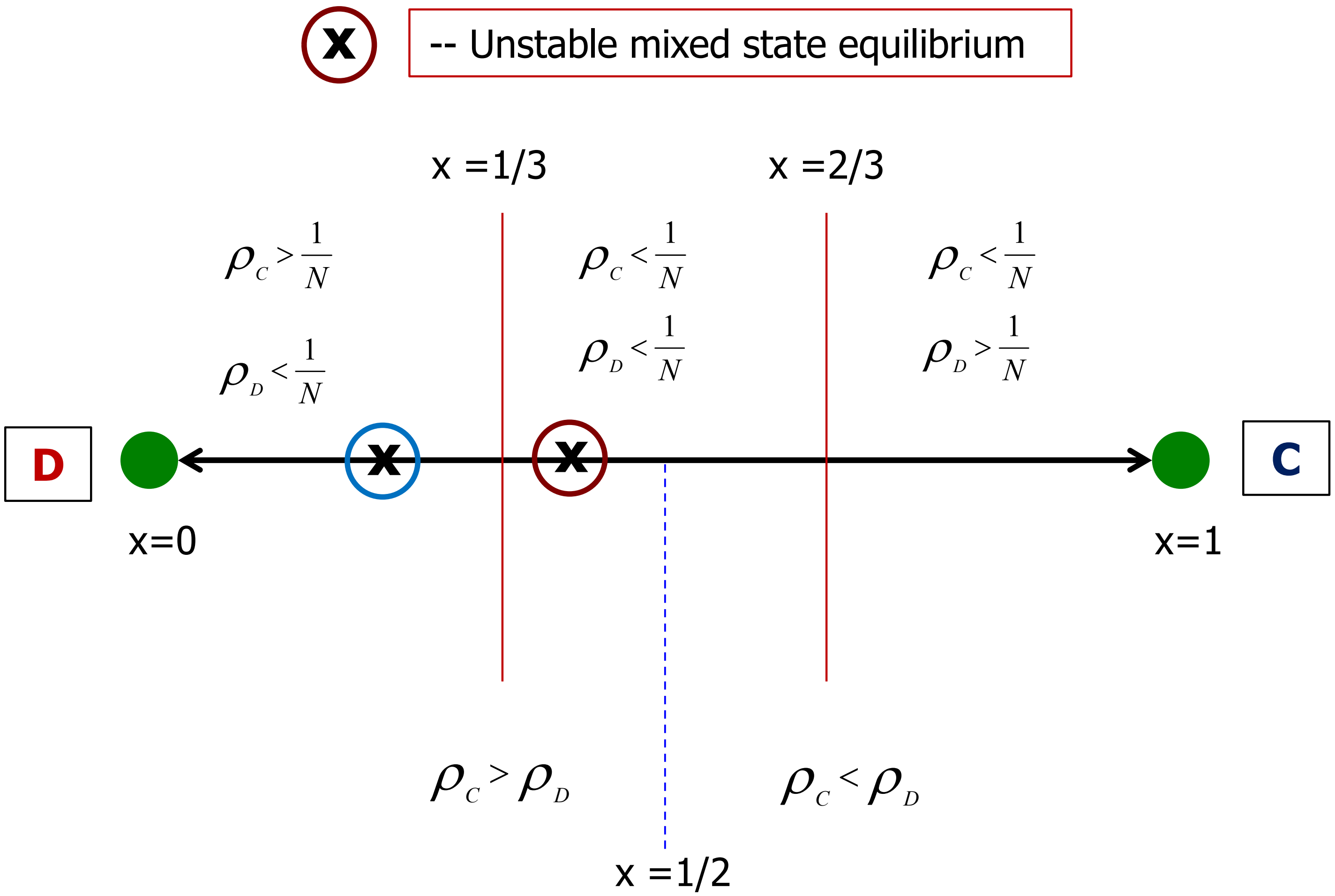
for $N \gg 1 \rightarrow \underline{a-d > c-b} \Rightarrow a+b > c+d$

$$\Rightarrow (a-c) > (d-b)$$

$$x^* = \frac{d-b}{(a-c)+(d-b)} < \frac{d-b}{(d-b)+(d-b)} = \frac{1}{2}$$

So, $x^* < \frac{1}{2}$

Fixation Probabilities and the 1/3 Law for w<<1 and N>>1



Risk Dominance: If both **C** and **D** is a strict Nash Equilibrium in the conventional sense i.e. if a>c and d>b then which strategy has a higher fixation probability ?

C is Risk Dominant if $\rho_C > \rho_D$ ➔ **a + b > c + d** when w<<1 and N>>1

D is Risk Dominant if $\rho_D > \rho_C$

A strategy is Risk Dominant if the total payoff for that strategy is larger than the total payoff for every other strategy.

The Risk Dominant strategy has a greater fixation probability in the limit w<<1 and N>>1

TFT can Invade ALLD in a Finite Population

For the Payoff Matrix

$$M = \begin{array}{c|cc} & \textbf{TFT} & \textbf{ALLD} \\ \textbf{TFT} & ma & b+(m-1)d \\ \textbf{ALLD} & C+(m-1)d & md \end{array}$$

No. of TFT players = **i**

No. of ALLD players = **N-i**

c>a>d>b

According to the *infinite population* analysis, for m > (c-d)/(a-d), both TFT and ALLD are an ESS and each strategy is stable against invasion by either strategy.

In finite populations, **TFT** can get fixed in the population even if $F_{TFT} < G_{ALLD}$ provided $\rho_{TFT} > \frac{1}{N}$

If F_i and G_i is the fitness of **i** **TFT** and (**N-i)** **ALLD** players,

$$F_i = \frac{ma(i-1)+(b+(m-1)d)(N-i)}{N-1} \qquad G_i = \frac{(c+(m-1)d)i+md(N-i-1)}{N-1}$$

$F_1 = F_{TFT} = b+(m-1)d$ and $G_1 = G_{ALLD} = (c+(m-1)d + md(N-2))/(N-1)$

For w→0 and fixed N, $\rho_{TFT} > \frac{1}{N}$ gives a lower bound on m: $m > \frac{c(N+1)+d(N-2)-b(2N-1)}{(a-d)(N-2)}$

When N=2, m>∞, When N=3: m>10 When N=4: m>6

For N>>1, lower bound on m: $m > \frac{c+d-2b}{(a-d)} \Rightarrow m > 3$ when a=3, b=0, c=5, d=1

$P_{TFT} > \frac{1}{N} \Rightarrow a(N-2) + b(2N-1) > c(N+1) + d(2N-4)$ $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

for TFT $\Rightarrow$

$ma(N-2) + (b+(m-1)d)(2N-1) >$

$(c+(m-1)d)(N+1) + md(2N-4)$ $\begin{pmatrix} ma & b+(m-1)d \\ c+(m-1)d & md \end{pmatrix}$

$\Rightarrow m(aN - 2a + d\cancel{2N} - d - dN - d - \cancel{2Nd} + 4d)$

$> -b(2N-1) + d(2N-1)$

$+ c(N+1) - d(N+1)$

$\Rightarrow m(aN - dN - 2a + 2d) > c(N+1) + d(N-2) - b(2N-1)$

$$\Rightarrow \boxed{m > \frac{c(N+1) + d(N-2) - b(2N-1)}{(a-d)(N-2)}}$$

for $N \gg 1$

$m > \dfrac{cN + dN - 2bN}{(a-d)N} \Rightarrow m > \dfrac{(c+d-2b)}{(a-d)}$

For $\begin{pmatrix} 3 & 0 \\ 5 & 1 \end{pmatrix} \Rightarrow \underline{m > 3}$

condition →

$$ma(N-2) + (b+(m-1)d)(2N-1) > (c+(m-1)d)(N+1) + md(2N-4)$$

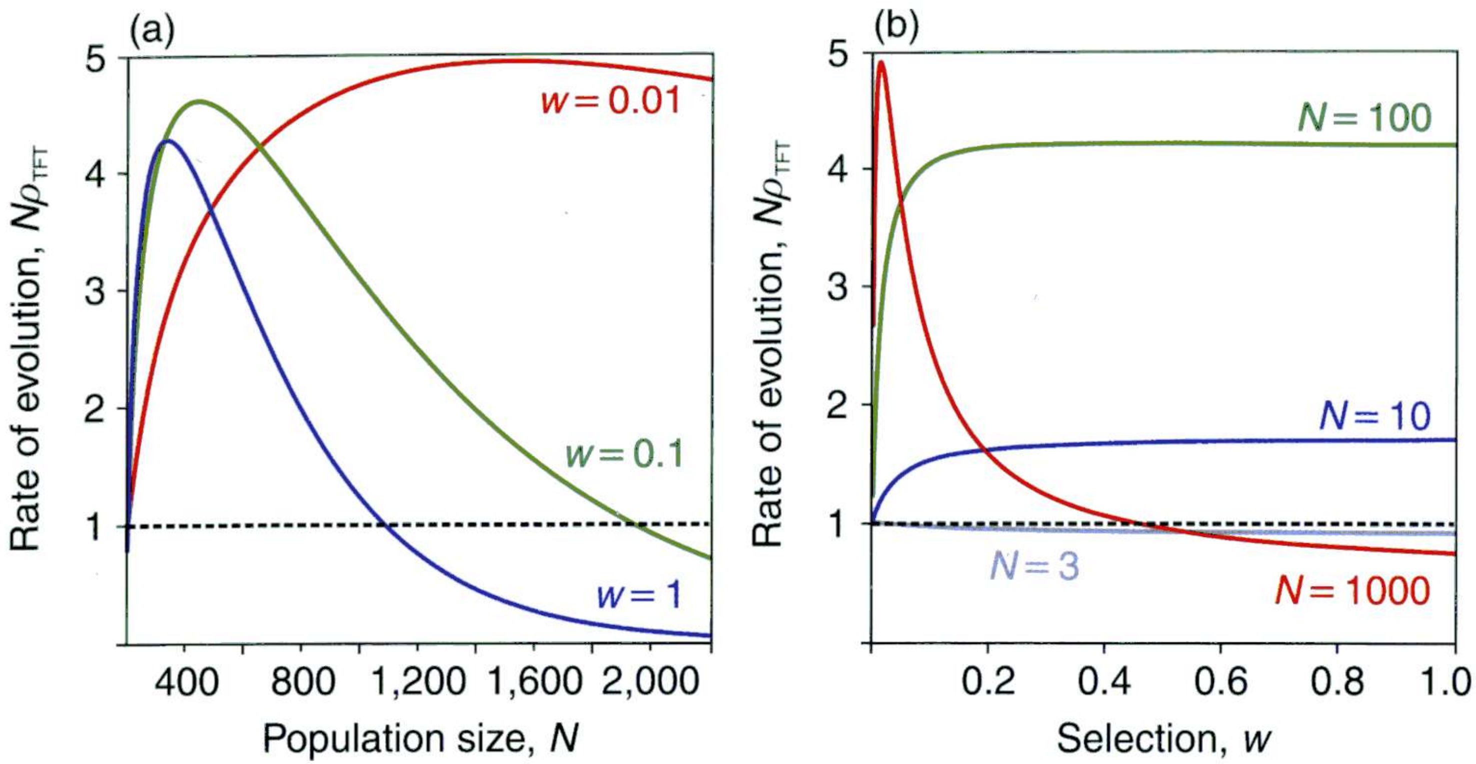
$$\Rightarrow N(ma + 2b + 2d(m-1) - c - (m-1)d - 2dm) > 2ma + b + (m-1)d + c + (m-1)d - 4md$$

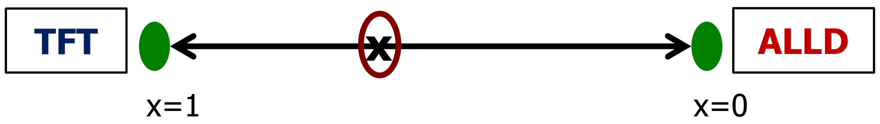
$$\Rightarrow N > \frac{2ma + b + c + d(m-1+m-1-4m)}{ma + 2b - c + d(\cancel{2m} - 2 - m + 1 - \cancel{2m})}$$

$$\Rightarrow \boxed{N > \frac{2ma + b + c - 2d(m+1)}{ma + 2b - c - d(m+1)}}$$

For fixed m, $\rho_{TFT} > \frac{1}{N}$ gives a *lower bound* on N: $N > \frac{2ma+b+c-2d(m+1)}{ma+2b-c-d(m+1)}$

For fixed m=10 : N > 3 (if a=3,b=0,c=5,d=1)





For very large N, stochastic fluctuations in #TFT are insufficient to take it from a freq of 1/N to x*

➔ Leads to an upper bound N_U (for moderate ***w***) on the population size for TFT to invade with probability > 1/N

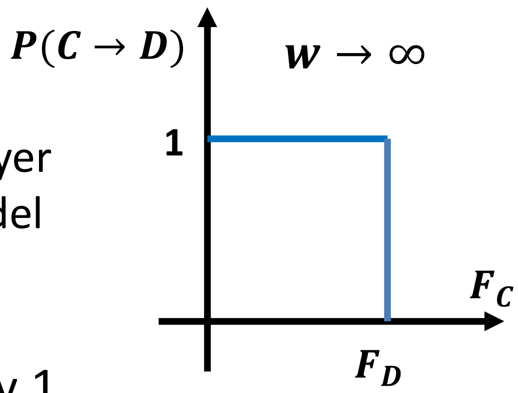
Summary

Criteria for evolutionary stability need to be revisited for finite populations

- Selfish strategies like ALLD that is an ESS in infinite population limit are guaranteed to be an ESS in finite population iff the population size (N) is above a threshold value.
- D cannot always prevent being invaded by C in finite populations
- TFT can invade ALLD if $N_c^{\min} < N < N_c^{\max}$
  - The lower threshold $N_c^{\min}$ is *analytically* determined in the limit w<<1
  - The upper threshold $N_c^{\max}$ is not revealed in the weak selection limit but emerges from numerical simulations when $w \leq 1$

Fixation Probability and Risk Dominance using the Fermi update rule

Payoff Comparison **method of population update**



Blume 1993, Szabo & Toke 1998
Traulsen, Pacheco, Nowak 2007

$$P(C \to D) = \frac{1}{1 + e^{-w(F_D - F_C)}}$$

C : Focal player
D : Role model

Limiting case: $w \to \infty$ If $F_D > F_C$, C is replaced by D with probability 1

If $F_D < F_C$, C is retained with probability 1

$$p_{i,i+1} \equiv \alpha_i = \left(\frac{i}{N}\right)\left(\frac{1}{1+e^{-w(F_i - G_i)}}\right)\left(\frac{N-i}{N}\right)$$

Probability of choosing C for reproduction

$$p_{i,i-1} \equiv \beta_i = \left(\frac{N-i}{N}\right)\left(\frac{1}{1+e^{-w(G_i - F_i)}}\right)\left(\frac{i}{N}\right)$$

Probability of choosing D for reproduction

$$\gamma_i = \frac{\beta_i}{\alpha_i} = e^{-w(F_i - G_i)} \qquad \frac{\rho_D}{\rho_C} = \prod_{i=1}^{N-1} \gamma_i = \prod_{i=1}^{N-1} e^{-w(F_i - G_i)} = e^{-w\sum_{i=1}^{N-1}(F_i - G_i)} = e^{-\frac{w}{2}X}$$

Valid for **all** selection strengths

$$X = (a+b-c-d)N - 2a + 2d > 0 \Rightarrow \rho_D < \rho_C$$

$$\boxed{N >> 1 : X > 0 \Rightarrow a + b > c + d}$$

Fixation Probability calculation using the Fermi update rule

$$\rho_C = \frac{1}{1+\sum_{k=1}^{N-1}\prod_{i=1}^{k}\gamma_i} = \frac{1}{1+\sum_{k=1}^{N-1}\prod_{i=1}^{k}e^{-w(F_i-G_i)}}$$

If **a - c = b - d** ⟹ Payoff difference is the same regardless of which strategy one of the players unilaterally switch from

$$\begin{array}{cc} & \begin{array}{cc} C & D \end{array} \\ \begin{array}{c} C \\ D \end{array} & \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{array}$$

$$\rho_C \equiv x_1 = \frac{1}{1+\sum_{k=1}^{N-1}e^{-wvk}} = \frac{1-e^{-wv}}{1-e^{-Nwv}}$$

➔ Invasion probability of **C**

$$x_i = \frac{1-e^{-wvi}}{1-e^{-wvN}}$$

compare

$$x_i = \frac{(1-\frac{1}{r^i})}{(1-\frac{1}{r^N})}$$

Equivalent to fixation probability in the constant selection case with $r = e^{wv}$

Fixation probability starting from a state with *i* C's

Pictorial representation of evolution using the Fermi update rule

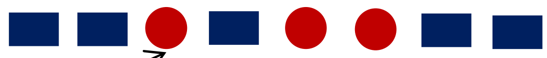
Time

C-type   D-type

Time=1

Pick a Random member

Suppose RN=3



Parent pop. Array (P)

D picked for reproduction

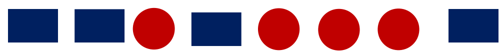
C picked for death at random

Compare Fitness of **D** and **C** for reproduction: Pick RN between 0&1. If **RN < P(C→D)**, is TRUE:

TRUE

D replaces **C** the seventh member of the Offspring pop. at t=1

Time=1



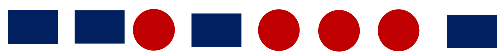
Offspring pop. Array (O)

D becomes the seventh member of the Offspring pop. at t=1

If RN > P(C→D) is FALSE: offspring same as that of parent

Offspring population at t=1 becomes the parent population t=2

Time=2



Parent pop. Array (P)

if **RN < P(C→D)**

Repeat above steps to generate the parent population t=3 and so on

$$P(C \to D) = \frac{1}{1 + e^{-w(F_D - F_C)}}$$

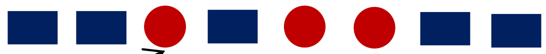
Note: By design, at most one change is happening in a time-step

Pictorial representation of evolution using the Fermi update rule: Whole population update

Time

C-type D-type

Time=1 Pick member i=1 and a random number (RN3) ε [1,N]



Parent pop. Array (P)

Suppose RN=3 **D** picked for payoff comparison

Compare Fitness of **D** and **C** for reproduction: Pick RN between 0&1. If **RN < P(C→D)**, is TRUE

RN < P(C→D): TRUE

D replaces **C** as the first member of the Offspring pop. at t=1

Time=1

Offspring pop. Array (O)

RN < P(C→D): FALSE

C is retained as the member of the Offspring pop. at t=1

→ Repeat for all members of the population

Offspring population at t=1 becomes the parent population t=2

Time=2

Parent pop. Array at t=2

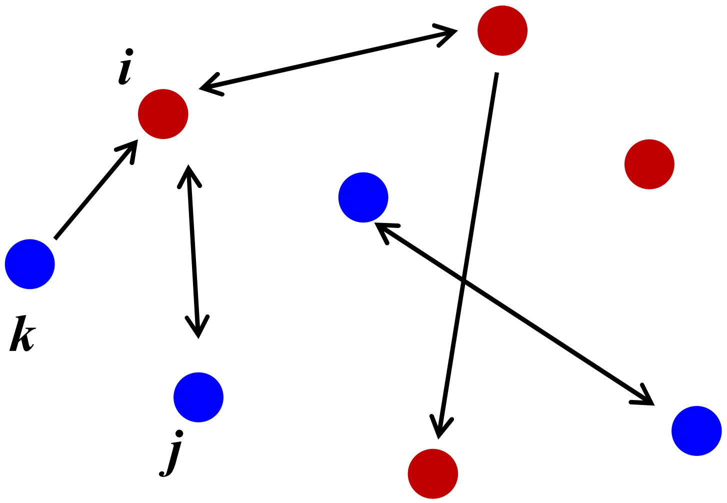
Repeat above steps to generate the parent population t=3 and so on

$$P(C \to D) = \frac{1}{1 + e^{-w(F_D - F_C)}}$$

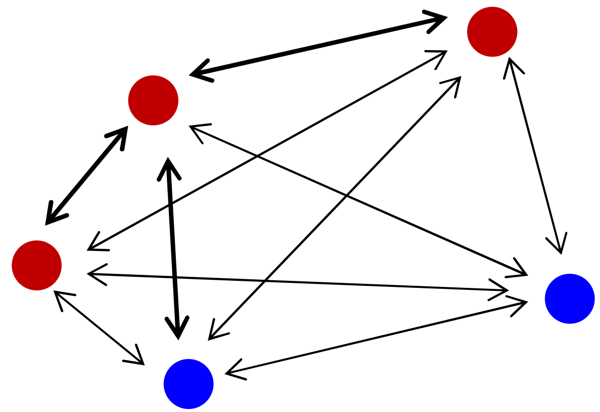
Evolutionary Graph Theory

Questions

- How does the fixation probability of a mutant change when the population is structured during only certain members of the population can replace others during the course of evolution.
- If a structured population is represented by a graph, with vertices representing members and edges representing interaction between corresponding members, is it possible to characterize ***all*** graphs that have the ***same*** evolutionary dynamics.
- Can certain structured populations increase the fixation probabilities of advantageous mutants ?
- Can certain structured populations eliminate the effect of selection ?



Structured Population



Unstructured Population

Not all vertices are connected by an edge

The edges can have different weights

i can replace j and j can replace i

k can replace i but i cannot replace k

There is an edge between ***any*** two vertices

All edges have the ***same*** weight

Consequences and Applications

Influence of networks on the spread of information

- Can cooperation spread more easily in certain networks?

  Reorganize organizational structure to facilitate cooperation?

- Consequences for cultural evolution

  Spread of innovation or misinformation in social networks?

- Consequences for disease spreading

  Design effective intervention techniques to minimize catastrophic consequences of epidemics?

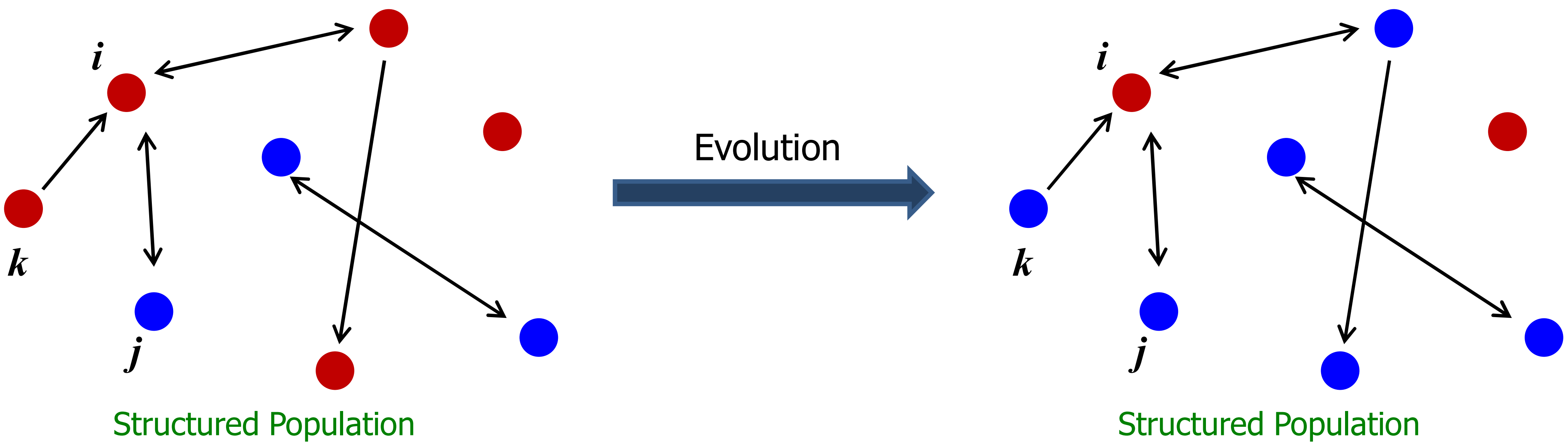
Formulating Evolution on Networks

A graph (network) can be completely specified by a stochastic matrix $W=[w_{ij}]$

$W=[w_{ij}]$ is an N x N stochastic matrix that determines the probability of replacing the j'th member of the population by the i'th member.

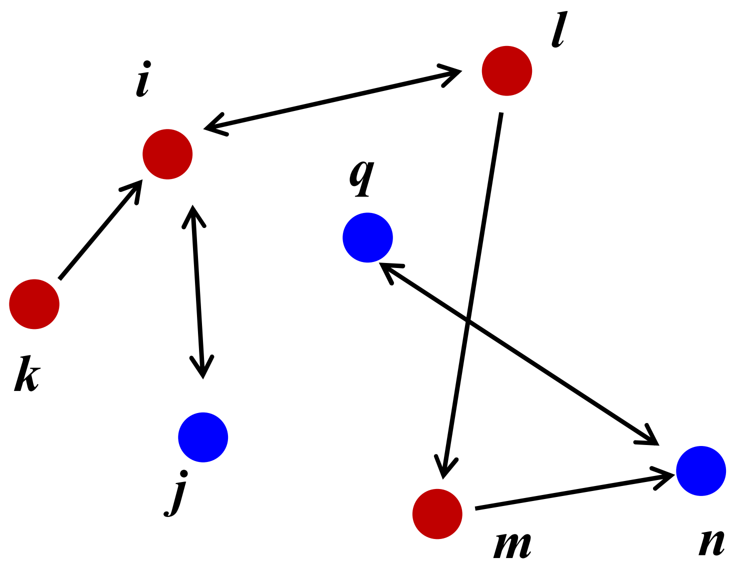
$w_{ij}=0$ if there is no directed edge *from* **i** *to* **j** ➔ offspring of i ***cannot*** replace j

$$\sum_{j=1}^{N} w_{ij} = 1$$ since the i'th member picked for reproduction has to replace someone



Population composition changes as the population evolves but the rules for replacement via the Moran process remain the same.

Representing network structure using a N x N stochastic matrix

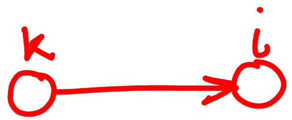


Structured Population

	i	j	k	l	m	n	q
i	0	1/2	0	1/2	0	0	0
j	1	0	0	0	0	0	0
k	1	0	0	0	0	0	0
l	1/2	0	0	0	1/2	0	0
m	0	0	0	0	0	1	0
n	0	0	0	0	0	0	1
q	0	0	0	0	0	1	0

$$\sum_{j=1}^{N} w_{ij} = 1$$

For a stochastic matrix, sum of the row elements must add up to 1



k can replace i
but i can't replace k

$W_{ij} \rightarrow$ Prob of i replace j

Initialize (I), Interact(I), Update (U), Repeat (R)

IIUR Structure of an agent-based simulation (ABS) of evolutionary games on networks

❖ Initial configuration: Specify **Initial** distribution of different strategies on a network

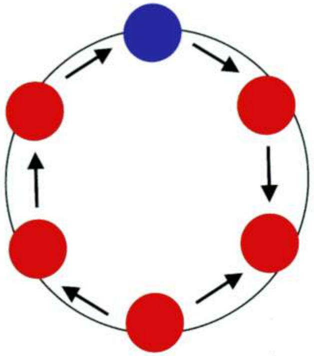
❖ Payoff calculation: **Interaction** and payoff calculation for every agent in the population

❖ Strategy update: **Update** strategies of agents using specified deterministic or stochastic update rules

synchronous ***asynchronous***

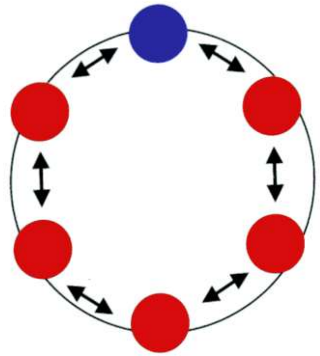
Fixation Probability of a mutant that arises in a structured population

$$W=\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



Directed Cycle

$$w=\begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{1}{2} & 0 \end{bmatrix}$$



Cycle

The i'th member can only be replaced by the member preceding it i.e the (i-1)th member.

Fitness of B (blue) = r
Fitness of A (red) = 1

Due to the nature of the structured population (only nearest neighbour replacements are allowed), there can be only one cluster of B's. Fragmentation of clusters into two or more sub-clusters is not possible.

$$\rho_B = \frac{1}{1+\sum_{k=1}^{N-1}\prod_{i=1}^{k}\gamma_i} = \frac{1-1/r}{1-1/r^N}$$

Fixation probability of B on a directed cycle is identical to the fixation probability of B in the Moran process (unstructured population)

Let $\#B = m$ and $\#A = N-m$; Let Relative fitness of B is r

Then, for B type agent, $P_{m,m-1} = \dfrac{1.f_A}{m.f_B + (N-m) f_A} = \dfrac{1}{mr + (N-m)}$

To increase the number of B $\rightarrow P_{m,m+1}$

$$P_{m,m+1} = \frac{1.f_B}{m.f_B + (N-m) f_A} = \frac{r}{mr + (N-m)}$$

$$\gamma_m = \frac{1}{r} \qquad \rho = \frac{1}{1 + \sum_{k=1}^{N-1} \prod_{m=1}^{k} \gamma_m} = \frac{1}{1 + \sum_{k=1}^{N-1} \frac{1}{r^k}}$$

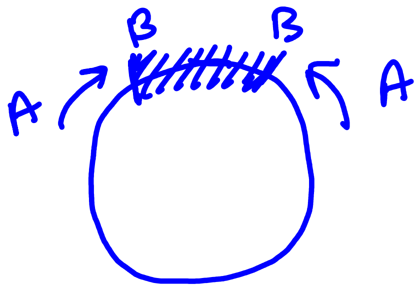
$$= \frac{1}{1 + \frac{1}{r} + \frac{1}{r^2} + \cdots \frac{1}{r^{N-1}}} = \frac{1 - 1/r^N}{1 - 1/r}$$

Bi Directed graph

The mutant always form a single contiguous cluster, and the symmetric edge weights on the boundaries of that cluster perfectly add up to one

Transition probability

- The A on the left has half chance of sending its offspring to the right (replacing B)
- The A on the right has half chance of sending its offspring to the left (replacing B)



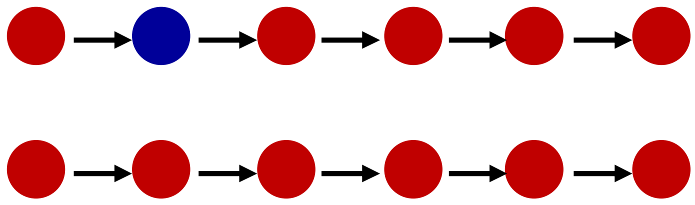
So, $P_{m,m-1} = 2 \times \frac{1}{2} \times \frac{1}{N-m+rm}$

Similarly,

$$P_{m,m+1} = 2 \times \frac{1}{2} \times \frac{r}{N-m+rm}$$

Fixation probability of a mutant randomly placed on a “Line” graph

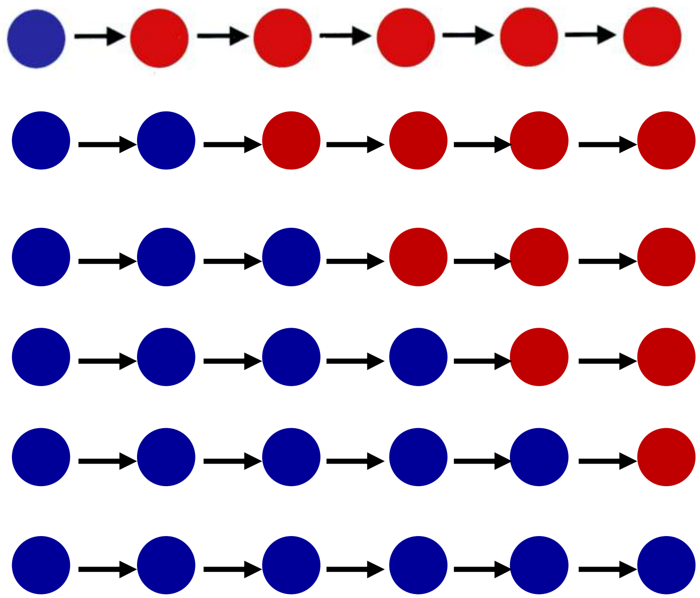
Line



Rules of Replacement: Every member is replaced only by the member preceding it. The last member replaces itself.

If the mutant B arises at any position other than the first position in the line, it will be replaced by A and become extinct.

Probability that B arises in any positions from i=2…N is (N-1)/N since there are N-1 such positions.



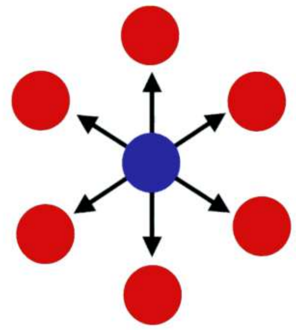
Probability that B appears in position 1: 1/N

The mutant B will definitely be fixed if it arises in position 1

Fixation probability of B: $\rho_B = \dfrac{1}{N}$

Fixation probability differs from the Moran process and is independent of the fitness of members.

Invasion probability of a mutant randomly placed on a “Burst” graph



Burst

Rules of Replacement: Every member is replaced only by the member at the centre with equal probability. The central member *cannot* be replaced by any other peripheral member or itself.

If the mutant B arises at any position other than the ***central*** position in the star, it will be replaced by the A at the centre and become extinct.

Probability that B arises in any positions from i=2…N is (N-1)/N since there are N-1 such positions.

Probability that B appears in the central position: 1/N

The mutant B will definitely be fixed if it appears at the centre of the star

Invasion probability of B: $\rho_B = \frac{1}{N}$

Invasion is *independent* of the fitness of members and equivalent to that of a neutral mutant in the Moran process.

Both the “Line” and ”Burst” graphs are suppressors of selection

Graphs which are suppressors or amplifiers of selection

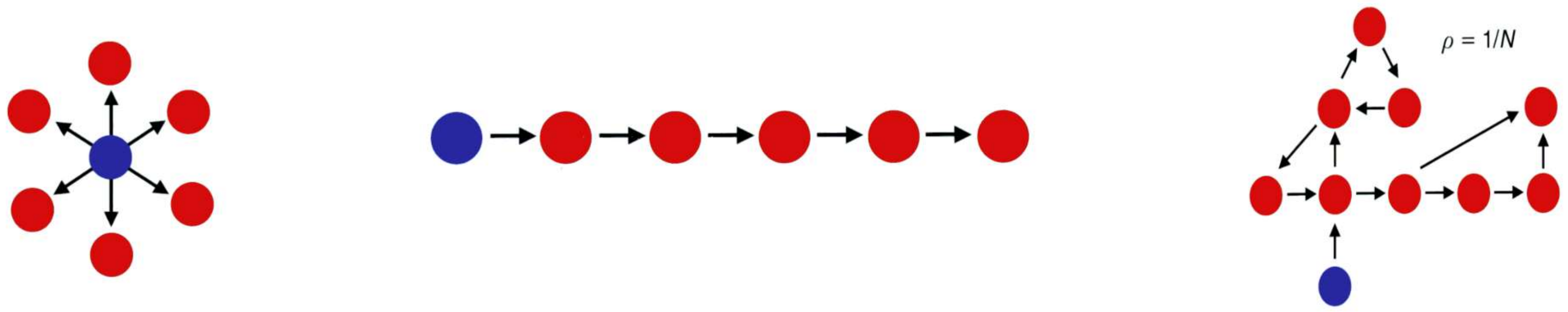
$$\rho_B = \frac{1-1/r}{1-1/r^N}$$ Invasion probability of a single mutant with a relative fitness ***r*** in a Moran process

If the fixation probability of a single mutant with a relative fitness r on the *structured graph* G is ρ_G

If $\rho_G > \rho_B$ when r>1 ➔ G is an amplifier of selection. G favours selection over drift

If $\rho_G < \rho_B$ when r>1 ➔ G is an suppressor of selection. G favours drift over selection

If $\rho_G = \frac{1}{N}$ when r>1 ➔ G is the strongest possible suppressor of selection.



All Graphs with a ***single root*** have the same invasion probability: $\rho_G = \frac{1}{N}$

$$\rho_M = \frac{1 - 1/r}{1 - \frac{1}{r^N}} \rightarrow$$ Fixation Probability of Moran Process

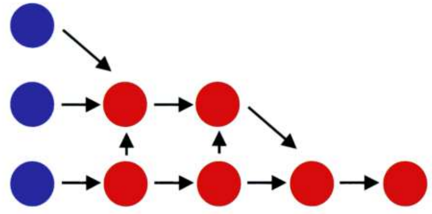
Mutation

- Advantagous $r > 1$
  - $\rho_G > \rho_M \rightarrow$ Amplifier of selection
  - $\rho_G < \rho_m \rightarrow$ Supressor of Selection
- Disadvantagous $r < 1$
  - $\rho_G > \rho_M \rightarrow$ Supressor of Selection
  - $\rho_G < \rho_m \rightarrow$ Amplifier of Selection

If $\rho_G = \frac{1}{N} \rightarrow$ Strongest possible suppressor of selection.

A *single* mutant B appearing in a graph with multiple roots can never get fixed since it cannot replace A located in one or more of the multiple roots.

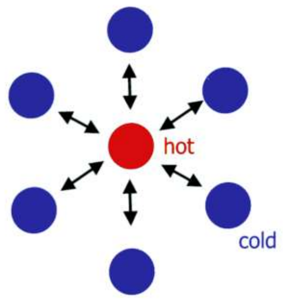
If a mutant appears at one of the roots, the lineage generated can never become extinct. Allows coexistence of both A and B in the population.



Graph with multiple roots

A root is a node with no edge pointing to it

Star Graph is an *amplifier* of selection: For r>1, a single mutant with relative fitness r has a higher fixation probability that is equivalent to the fixation probability with relative fitness r^2 in a Moran process (unstructured graph) in the limit of ***large*** N



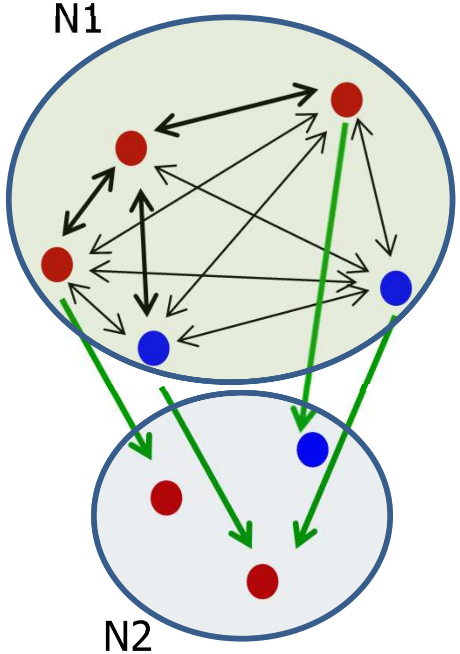
Star

$$\rho_{B,star} = \frac{1-1/r^2}{1-1/r^{2N}}$$

A compartmentalized graph which has a complete sub-graph is an amplifier of selection: For r>1, a single mutant with relative fitness r has a larger fixation probability than the corresponding fixation probability in a Moran process (unstructured graph).

SS1

$N=N_1+N_2$; where N1 is the size of the complete (unstructured) sub-graph and N2 is the size of the second component.



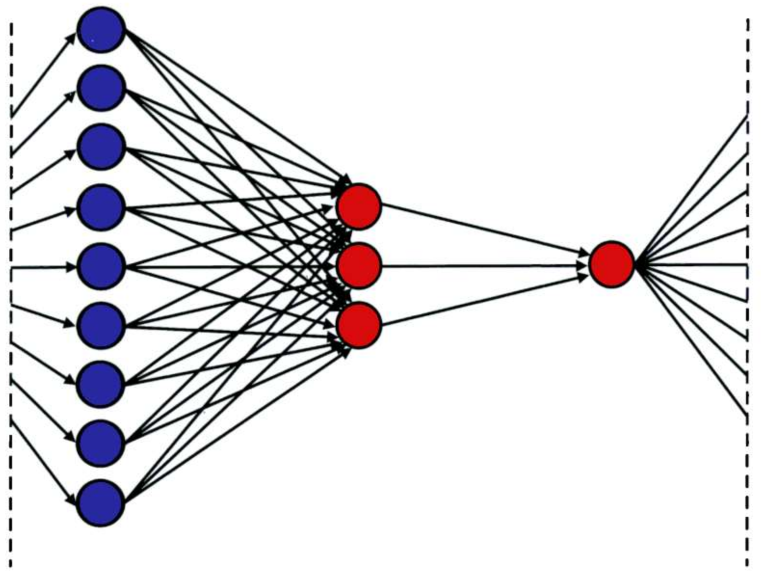
$$\rho_B = \frac{1}{1+\sum_{j=1}^{N_1-1}\prod_{k=1}^{j}\gamma_k} = \frac{1-1/r}{1-1/r^{N_1}}$$

Slide 133

SS1 Fixation of B can occur only if B is a node in the complete sub-graph of size N1 since only nodes in N1 can replace nodes in N2 and not vice versa, the sum should be only over nodes in the complete subgraph of size N1

Supratim Sengupta, 26-02-2020

The Funnel graph is a strong amplifier of selection



The Funnel graph has **k+1** layers labelled **j=0,1,...k**

The zeroth layer has just one vertex and the **layer j has m^j vertices**

Edges originating from vertices in layer j lead to vertices in layer j-1

Edges originating from the single vertex in the layer j=0 lead back to vertices in the layer j=k

In the limit of ***large*** k and ***large*** m,

$\rho_G \to 1$ **for r>1**

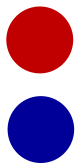
$\rho_G \to 0$ **for r<1**

Reference: Lieberman, Hauert, Nowak; Evolutionary Dynamics on Graphs, Nature 2005

Key Question

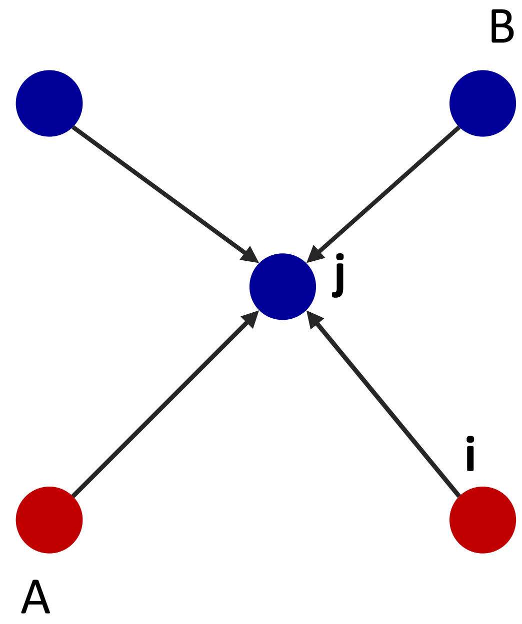
Is there a criterion for deciding whether the underlying population structure *changes* the *fixation probability* when compared to the *well-mixed* population scenario?

$p_{m,m-1}$ calculation



A #A=N-m

B #B=m

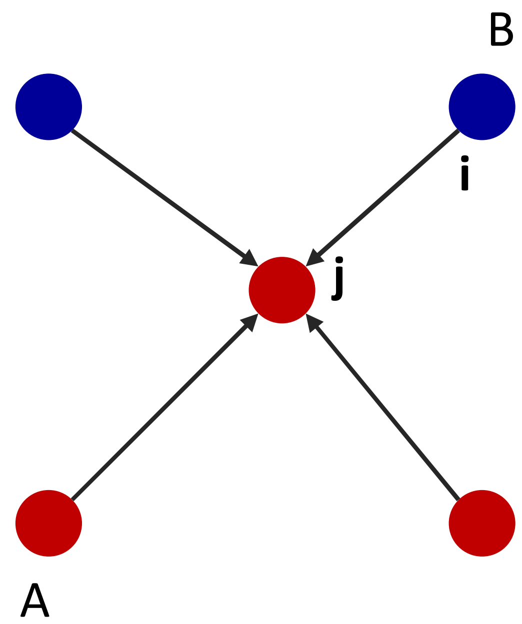


m➔m-1 when the j’th node picked for death is **B** and the i’th node picked for reproduction in **A**

possible ways in which A can replace B :

$$\sum_{i,j} w_{ij}(1-\nu_i)\nu_j$$ = # edges directed from A ➔ B

$p_{m,m+1}$ calculation



m➔m+1 when the j’th node picked for death is **A** and the i’th node picked for reproduction in **B**

possible ways in which B can replace A :

$$\sum_{i,j} w_{ij}(1-\nu_j)\nu_i$$ = # edges directed from B ➔ A

The Isothermal Theorem

$$T_j = \sum_{i=1}^{N} w_{ij}$$

T_j : Sum of all weights that lead to the vertex.
A vertex with high temperature is replaced more frequently than a vertex with low temperature

Isothermal Graph: All vertices have the *same* temperature ➔ $T_j = \sum_{i=1}^{N} w_{ij} = const = 1$

Isothermal Theorem: The fixation probability of a single mutant on a graph G is equivalent to the fixation probability on an unstructured graph (Moran process) iff G is an isothermal graph.

$v_i = 1$ ➔ vertex i is occupied by B, $v_i = 0$ ➔ vertex i is occupied by A

$m = \sum_{i=1}^{m} v_i$ ➔ No. of B's in the population

$$p_{m,m+1} = \frac{r \sum_i \sum_j w_{ij} v_i (1 - v_j)}{rm + N - m} \qquad p_{m,m-1} = \frac{\sum_i \sum_j w_{ij} v_j (1 - v_i)}{rm + N - m}$$

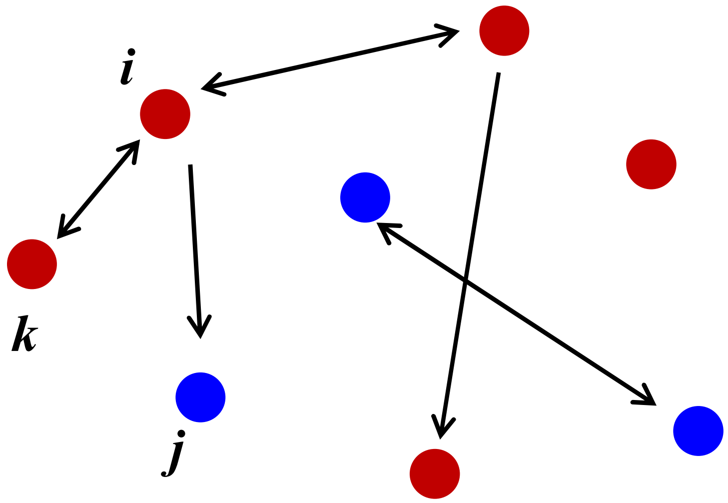
Fixation probability is the same as the Moran process if $\dfrac{p_{m,m-1}}{p_{m,m+1}} = \dfrac{1}{r}$ ➔

$$\sum_i \sum_j w_{ij} v_j (1 - v_i) = \sum_i \sum_j w_{ij} v_i (1 - v_j) \quad ➔ \quad \sum_i \sum_j w_{ji} v_i (1 - v_j) = \sum_i \sum_j w_{ij} v_i (1 - v_j)$$

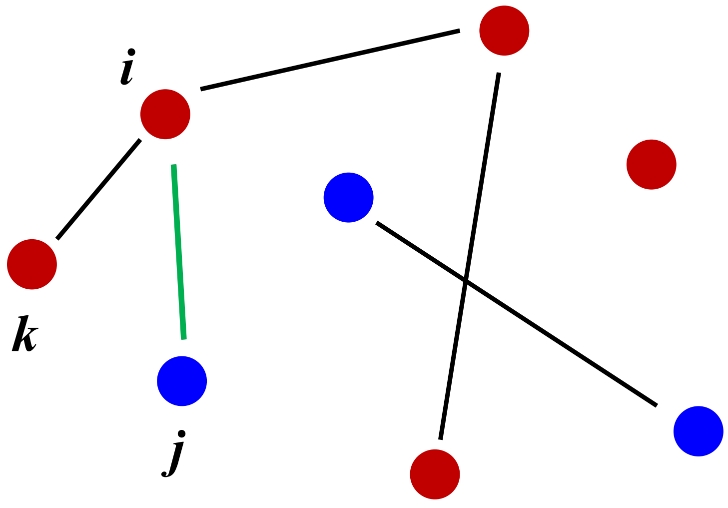
The above equation is valid for all i. Specifically for a particular i=k such that $v_k = 1, v_i = 0$ for all $i \neq k$

➔ $\sum_j w_{jk} = \sum_j w_{kj}$ Since $\sum_{j=1}^{N} w_{kj} = 1$ ➔ $T_k \equiv \sum_{j=1}^{N} w_{jk} = 1$

Alternative way to represent evolutionary dynamics on Graphs



Structured Population



Structured Population

Instead of picking a vertex for reproduction and another vertex (which it can replace) for death

Pick an Edge (ij) with probability proportional to (w_{ij} x f_i)

f_i - fitness of the member at vertex i

Arrows are no longer necessary: weight w_{ij} contains information about which member replaces which

W=[w_{ij}] need not be a stochastic matrix. w_{ij} can be any *non-negative* number $$\sum_{j=1}^{N} w_{ij} \neq 1$$

Fixation probability is same as that of a Moran Process in unstructured populations if the graph is a circulation i.e.

$$\sum_j w_{jk} = \sum_j w_{kj}$$

Every *Isothermal* graph is a *Circulation* but not every Circulation is *Isothermal*

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LETTERS

A simple rule for the evolution of cooperation on graphs and social networks

Hisashi Ohtsuki[1,2], Christoph Hauert[2], Erez Lieberman[2,3] & Martin A. Nowak[2]

A fundamental aspect of all biological systems is cooperation. Cooperative interactions are required for many levels of biological organization ranging from single cells to groups of animals[1–4]. Human society is based to a large extent on mechanisms that promote cooperation[5–7]. It is well known that in unstructured populations, natural selection favours defectors over cooperators. There is much current interest, however, in studying evolutionary games in structured populations and on graphs[8–17]. These efforts recognize the fact that who-meets-whom is not random, but determined by spatial relationships or social networks[18–24]. Here we describe a surprisingly simple rule that is a good approximation for all graphs that we have analysed, including cycles, spatial lattices, random regular graphs, random graphs and scale-free networks[25,26]: natural selection favours cooperation, if the benefit of the altruistic act, *b*, divided by the cost, *c*, exceeds the average number of neighbours, *k*, which means $b/c > k$. In this case, cooperation can evolve as a consequence of 'social viscosity' even in the absence of reputation effects or strategic complexity.

A cooperator is someone who pays a cost, *c*, for another individual to receive a benefit, *b*. A defector pays no cost and does not distribute any benefits. In evolutionary biology, cost and benefit are measured in terms of fitness. Reproduction can be genetic or cultural. In the latter case, the strategy of someone who does well is imitated by others. In an unstructured population, where all individuals are equally likely to interact with each other, defectors have a higher average payoff than unconditional cooperators. Therefore, natural selection increases the relative abundance of defectors and drives cooperators to extinction. These evolutionary dynamics hold for the deterministic setting of the replicator equation[27,28] and for stochastic game dynamics of finite populations[29].

In our model, the players of an evolutionary game occupy the vertices of a graph. The edges denote links between individuals in terms of game dynamical interaction and biological reproduction. We assume that the graph is fixed for the duration of the evolutionary dynamics. Consider a population of N individuals consisting of cooperators and defectors. A cooperator helps all individuals to whom it is connected. If a cooperator is connected to k other individuals and i of those are cooperators, then its payoff is $bi - ck$. A defector does not provide any help, and therefore has no costs, but it can receive the benefit from neighbouring cooperators. If a defector is connected to j cooperators, then its payoff is bj.

The fitness of an individual is given by a constant term, denoting the baseline fitness, plus the payoff that arises from the game. Strong selection means that the payoff is large compared to the baseline fitness; weak selection means the payoff is small compared to the baseline fitness. The idea behind weak selection is that many different factors contribute to the overall fitness of an individual, and the game under consideration is just one of those factors.

At first, we will study the following update rule for evolutionary dynamics (Fig. 1): in each time step, a random individual is chosen to die, and the neighbors compete for the empty site proportional to their fitness. We call this mechanism 'death–birth' updating, because it involves a death event followed by a birth. Later we will investigate other update mechanisms.

Let us explore whether natural selection can favour cooperation on certain graphs. To do this, we need to calculate the probability that a single cooperator starting in a random position turns the whole population from defection to cooperation. If selection neither favours nor opposes cooperation, then this probability is $1/N$, which is the fixation probability of a neutral mutant. If the fixation

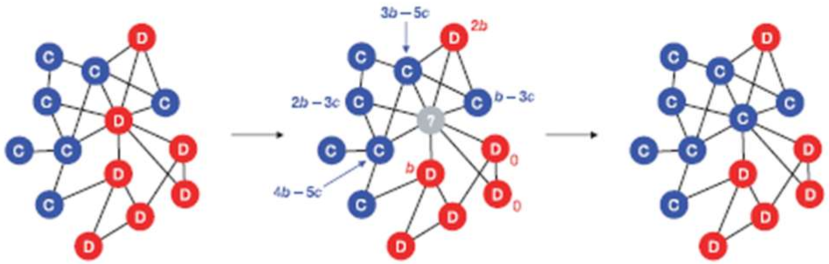


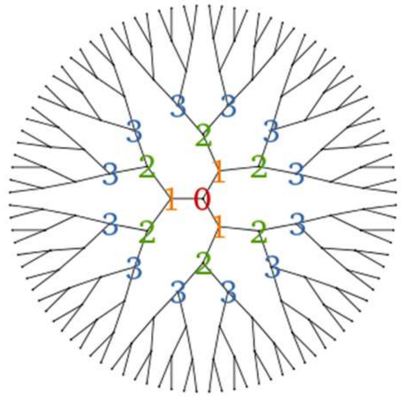
Figure 1 | The rules of the game. Each individual occupies the vertex of a graph and derives a payoff, *P*, from interactions with adjacent individuals. A cooperator (blue) pays a cost, *c*, for each neighbour to receive a benefit, *b*. A defector (red) pays no cost and provides no benefit. The fitness of a player is given by $1 - w + wP$, where w measures the intensity of selection. Strong selection means $w = 1$. Weak selection means $w \ll 1$. For 'death–birth' updating, at each time step, a random individual is chosen to die (grey); subsequently the neighbours compete for the empty site in proportion to their fitness. In this example, the central, vacated vertex will change from a defector to a cooperator with a probability $F_C/(F_C + F_D)$, where the total fitness of all adjacent cooperators and defectors is $F_C = 4(1 - w) + (10b - 16c)w$ and $F_D = 4(1 - w) + 3bw$, respectively.

[1]Department of Biology, Kyushu University, Fukuoka 812-8581, Japan. [2]Program for Evolutionary Dynamics, Department of Organismic and Evolutionary Biology, Department of Mathematics, [3]Department of Applied Mathematics, Harvard University, Cambridge, Massachusetts 02138, USA.

Constraints on the Theoretical Formulation

- Caley Tree/Bethe Lattice
  - *Regular* graph with each node having **k** neighbours
  - Graph does not have any loops

Image Source: Wikipedia



Caley Tree/Bethe Lattice with k=3

- Theoretical analysis valid for
  - N>>k
  - Weak selection limit holds i.e. w<<1 when separation of time-scales is possible
  - Uses the pair approximation which is valid only for Bethe lattices i.e. graphs without any loops.

Pair Approximation ➔ frequencies of larger clusters obtained from pair frequencies

Relations between stochastic variables for evolution of cooperation on graphs

$$p_A + p_B = 1 \Rightarrow p_B = 1 - p_A$$

$$q_{A|B} + q_{B|B} = 1; q_{B|A} + q_{A|A} = 1$$

$$p_{AA} = q_{A|A} p_A$$

$$p_{AB} = q_{A|B} p_B = q_{A|B}(1 - p_A)$$

$$p_{BB} = q_{B|B} p_B = (1 - q_{A|B})(1 - p_A)$$

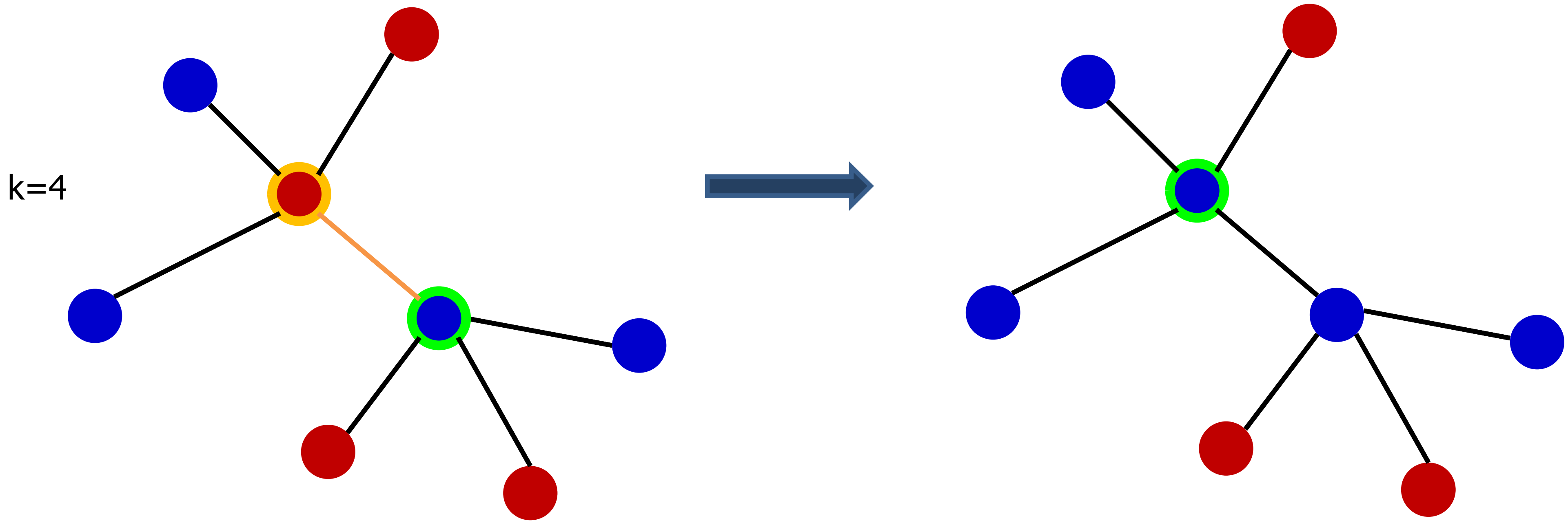
$$p_{BA} = p_{AB} \Rightarrow q_{B|A} p_A = q_{A|B} p_B \Rightarrow q_{A|B} = (1 - q_{A|A})(\frac{p_A}{1 - p_A})$$

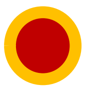
Only 2 of the 6 stochastic variables are *independent*: $p_A; p_{AA}$ or $p_A; q_{AA}$

AIM: Obtain dynamical equations in terms of these variables and solve them under certain approximations to obtain the condition for the fixation probability of A (ρ_A) > 1/N

Key approx.: *Pair frequencies equilibrate in a much faster time-scale than individual frequencies* in the weak selection limit (w<<1)

Illustration of the Death-Birth Process : Updating a B-player




Focal player selected for death is B

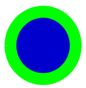
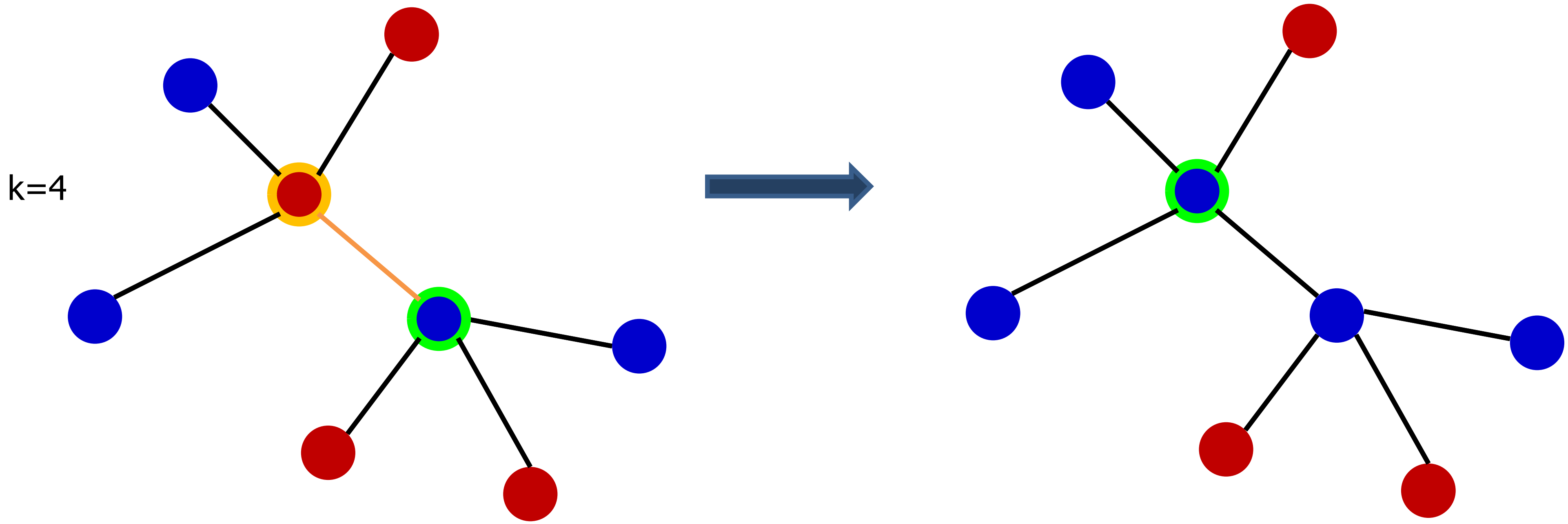
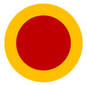
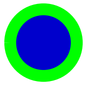
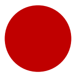

A-neighbour of focal B selected to replace B

Illustration of the Death-Birth Process : Updating a B-player



 Focal player selected for death is B

 A-neighbour of focal B selected to replace B has payoff: $P_A = b + 3q_{A|A}a + 3q_{B|A}b$

 B-neighbour of focal B *not* selected to replace B has payoff: $P_B = d + 3q_{B|B}d + 3q_{A|B}c$

$f_A = 1 - w + wP_A$: Fitness of A-neighbour of focal B

$f_B = 1 - w + wP_B$: Fitness of B-neighbour of focal B

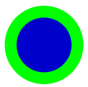
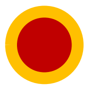
Probability that  replaces the focal  : $\dfrac{k_A f_A}{k_A f_A + k_B f_B}$

Illustration of the Death-Birth Process : Updating a B-player

$$p_B \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|B}}^{k_A} {q_{B|B}}^{k_B}\right] \frac{k_A f_A}{(k_A f_A + k_B f_B)}$$

Total fitness of $\boldsymbol{k_A}$ A-neighbours

Total fitness of all **k**-neighbours of B

Probability of choosing B for death

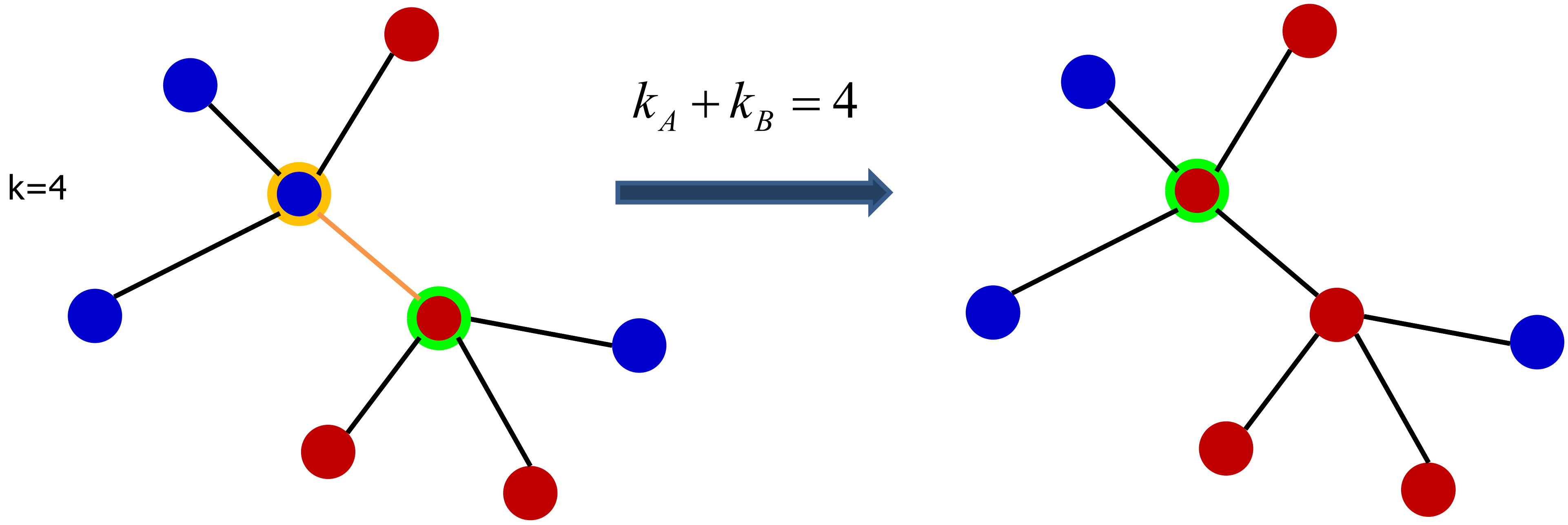
Number of ways of picking B having $\boldsymbol{k_A}$ A-neighbours and $\boldsymbol{k_B}$ B-neighbours

A is selected to replace B with probability proportional to the *relative* fitness of $\boldsymbol{k_A}$ A-neighbours of the focal B

$$Prob.\left(\Delta p_A = \frac{1}{N}\right) = p_B \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|B}}^{k_A} {q_{B|B}}^{k_B}\right] \left(\frac{k_A f_A}{k_A f_A + k_B f_B}\right)$$

$$Prob.\left(\Delta p_A = \frac{2k_A}{Nk}\right) = p_B \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|B}}^{k_A} {q_{B|B}}^{k_B}\right] \left(\frac{k_A f_A}{k_A f_A + k_B f_B}\right)$$

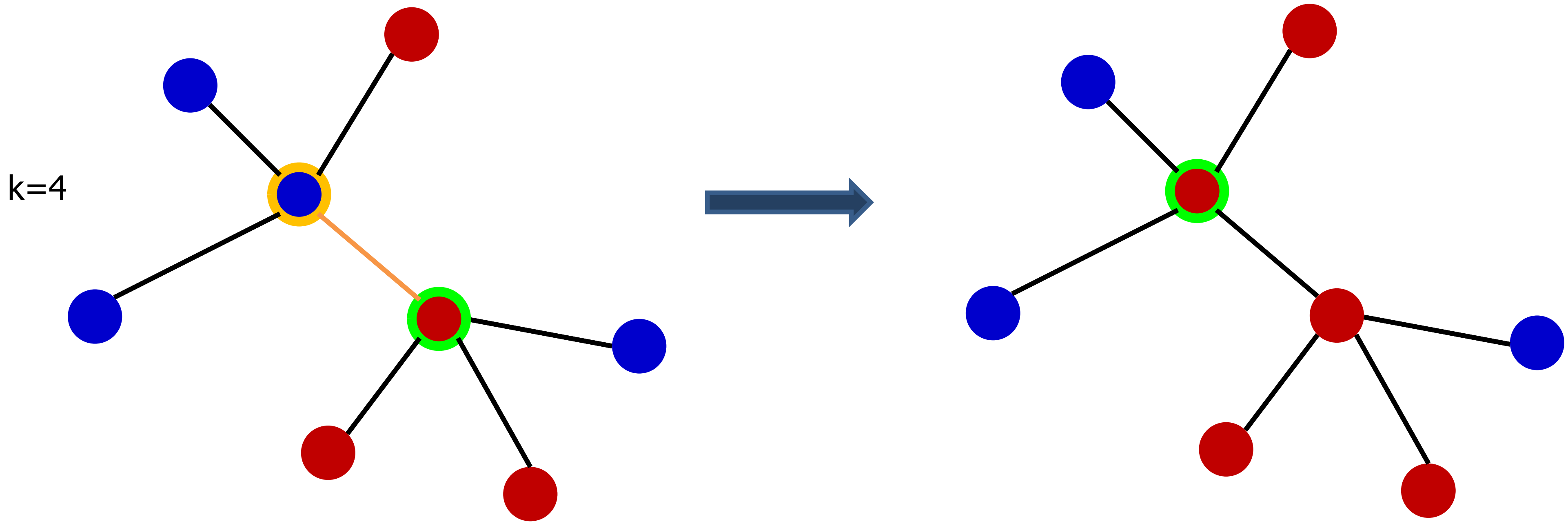
Illustration of the Death-Birth Process : Updating an A-player



Focal player selected for death is A

B-neighbour of focal A

Illustration of the Death-Birth Process : Updating an A-player



Focal player selected for death is A

A-neighbour of focal A *not* selected to replace A has payoff: $P_A = a + 3q_{A|A}a + 3q_{B|A}b$

B-neighbour of focal A selected to replace A has payoff: $P_B = c + 3q_{B|B}d + 3q_{A|B}c$

$g_A = 1 - w + wP_A$: Fitness of A-neighbour of focal A $g_B = 1 - w + wP_B$: Fitness of B-neighbour of focal A

Probability that replaces the focal : $\dfrac{k_B f_B}{k_A f_A + k_B f_B}\ ; k_A + k_B = 4$

Illustration of the Death-Birth Process : Updating an A-player

$$p_A \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|A}}^{k_A} {q_{B|A}}^{k_B}\right] \frac{k_B g_B}{(k_A g_A + k_B g_B)}$$

Total fitness of $\boldsymbol{k_B}$ B-neighbours

Total fitness of all **k**-neighbours of A

Probability of choosing A for death

Number of ways of picking A having $\boldsymbol{k_A}$ A-neighbours and $\boldsymbol{k_B}$ B-neighbours

B is selected to replace A with probability proportional to the *relative* fitness of $\boldsymbol{k_B}$ B-neighbours of the focal A

$$Prob.\left(\Delta p_A = -\frac{1}{N}\right) = p_A \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|A}}^{k_A} {q_{B|A}}^{k_B}\right] \left(\frac{k_B g_B}{k_A g_A + k_B g_B}\right)$$

$$Prob.\left(\Delta p_{AA} = -\frac{2k_A}{Nk}\right) = p_A \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|A}}^{k_A} {q_{B|A}}^{k_B}\right] \left(\frac{k_B g_B}{k_A g_A + k_B g_B}\right)$$

Rate of Change of Frequency of A

$$Prob.\left(\Delta p_A = \frac{1}{N}\right) = p_B \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|B}}^{k_A} {q_{B|B}}^{k_B}\right] \left(\frac{k_A f_A}{k_A f_A + k_B f_B}\right)$$

$$Prob.\left(\Delta p_A = -\frac{1}{N}\right) = p_A \sum_{\substack{k_A=0 \\ k_A+k_B=k}}^{k} \left[\frac{k!}{k_A!\,k_B!} {q_{A|A}}^{k_A} {q_{B|A}}^{k_B}\right] \left(\frac{k_B g_B}{k_A g_A + k_B g_B}\right)$$

$$\frac{d}{dt} p_A = \frac{1}{N} Prob.\left(\Delta p_A = \frac{1}{N}\right) - \frac{1}{N} Prob.\left(\Delta p_A = -\frac{1}{N}\right)$$

$$\frac{d}{dt} p_A = w\frac{k-1}{N} p_{AB}(aI_a + bI_b - cI_c - dI_d) + O(w^2)$$

$$I_a = \frac{k-1}{k} q_{A|A}\left(q_{A|A} + q_{B|B}\right) + \frac{1}{k} q_{A|A}$$

$$I_b = \frac{k-1}{k} q_{B|A}\left(q_{A|A} + q_{B|B}\right) + \frac{1}{k} q_{B|B}$$

$$I_c = \frac{k-1}{k} q_{A|B}\left(q_{A|A} + q_{B|B}\right) + \frac{1}{k} q_{A|A}$$

$$I_d = \frac{k-1}{k} q_{B|B}\left(q_{A|A} + q_{B|B}\right) + \frac{1}{k} q_{B|A}$$

Death-Birth (DB) updating: $\rho_C > \frac{1}{N} > \rho_D \quad \Rightarrow \quad \frac{b}{c} > k$

Backward Kolmogorov Equation and Calculation of Fixation Probability

$\rho_A(p,t|p_0)$: Probability of finding the population in a state where the frequency of A is **p** at time '**t**', given that it was $\mathbf{p_0}$ at time **t=0**

$g(p,\varepsilon;dt)$: Prob. that frequency of A from **p** at time '**t**' to **p+ε** at time **t+dt**

$$\rho_A\left(p,t+dt|p_0\right)=\int \mathrm{d}\varepsilon g\left(p_0,\varepsilon;dt\right)\rho_A\left(p,t|p_0+\varepsilon\right)$$

$$\frac{\partial\rho_A\left(p,t|p_0\right)}{\partial t}=M\left(p_0\right)\frac{\partial\rho_A\left(p,t|p_0\right)}{\partial p_0}+\frac{V\left(p_0\right)}{2}\frac{\partial^2\rho_A\left(p,t|p_0\right)}{\partial p_0^2}$$

Mean: $M\left(p_0\right)dt=\int \varepsilon g\left(p_0,\varepsilon;dt\right)\mathrm{d}\varepsilon$

Second Moment: $V\left(p_0\right)dt=\int \varepsilon^2 g\left(p_0,\varepsilon;dt\right)\mathrm{d}\varepsilon$

$$\frac{\partial\rho_A\left(p,t|p_0\right)}{\partial t}=0$$

at equilibrium

$$M\left(p_0\right)\frac{\partial\rho_A\left(p,t|p_0\right)}{\partial p_0}+\frac{V\left(p_0\right)}{2}\frac{\partial^2\rho_A\left(p,t|p_0\right)}{\partial p_0^2}=0$$

Fixation Probability

$$\rho_A\left(p=1,t=\infty|p_0=\frac{1}{N}\right)\equiv\rho_A\left(p_0=\frac{1}{N}\right)$$

Condition for Spread of Cooperation on Networks

Ohtsuki *et al.* A simple rule for evolution of cooperation on graphs and social networks; Nature 441 (2006) 502

Imitation (IM) updating: Payoff of focal individual being updated also matters.

Focal individual (F) compares her fitness with her neighbours.

F retains her strategy if f_F > f_neighbour (neighbour is randomly selected)

F imitates neighbour with probability proportional to neighbour's fitness if f_F<f_neighbour

f_F = 1-w+w(k_Ac+k_Bd) ; f_F: fitness of Focal B-player with k_A A-neighbours & k_B B-neighbours

Probability that a focal B-player adopts the strategy of an A-neighbour: $\dfrac{k_A f_A}{k_A f_A + k_B f_B + f_F}$

Compare with

Probability that a focal B is replaced by an A-neighbour in Death-Birth updating: $\dfrac{k_A f_A}{k_A f_A + k_B f_B}$

g_F = 1-w+w(k_Aa+k_Bb)   g_F: fitness of Focal A-player with k_A A-neighbours & k_B B-neighbours

Probability that a focal A-player adopts the strategy of a B-neighbour: $\dfrac{k_B g_B}{k_A g_A + k_B g_B + g_F}$

Ratio of Benefit to Cost of cooperation determines whether selection favours spread of cooperation and fixation of cooperators

$$\rho_C > \frac{1}{N} > \rho_D \quad \Rightarrow \quad \frac{b}{c} > k + 2$$

Condition for Spread of Cooperation on Networks

Ohtsuki *et al.* A simple rule for evolution of cooperation on graphs and social networks; Nature 441 (2006) 502

Birth-Death (BD) updating:

Probability that an A-player with k_A A-neighbours and k_B B-neighbours is selected for reproduction

$$\left[p_A \frac{k!}{k_A!k_B!} q_{A|A}{}^{k_A} q_{B|A}{}^{k_B}\right]\left[1-w+w(k_A a+k_B b)\right]$$

Frequency of the A-player with k_A A-neighbours & k_B B-neighbours

Fitness of selected A-player

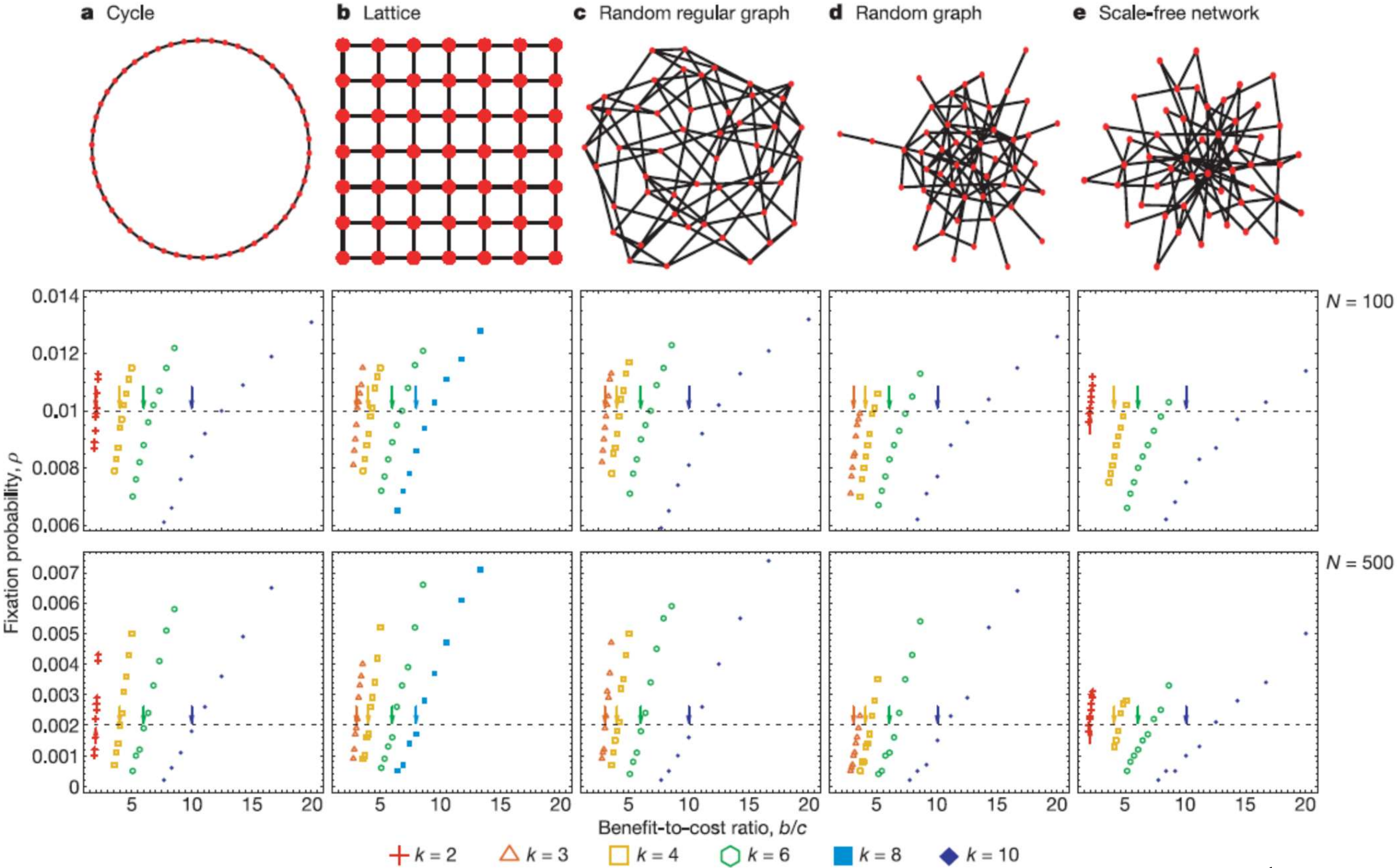
Probability that a B-player with k_A A-neighbours and k_B B-neighbours is selected for reproduction

$$\left[p_B \frac{k!}{k_A!k_B!} q_{A|B}{}^{k_A} q_{B|B}{}^{k_B}\right]\left[1-w+w(k_A c+k_B d)\right]$$

$$\rho_D > \frac{1}{N} > \rho_C \Longrightarrow$$ Selection never favours fixation of cooperators

Condition for Spread of Cooperation on Networks

Ohtsuki *et al.* A simple rule for evolution of cooperation on graphs and social networks; Nature (2006)

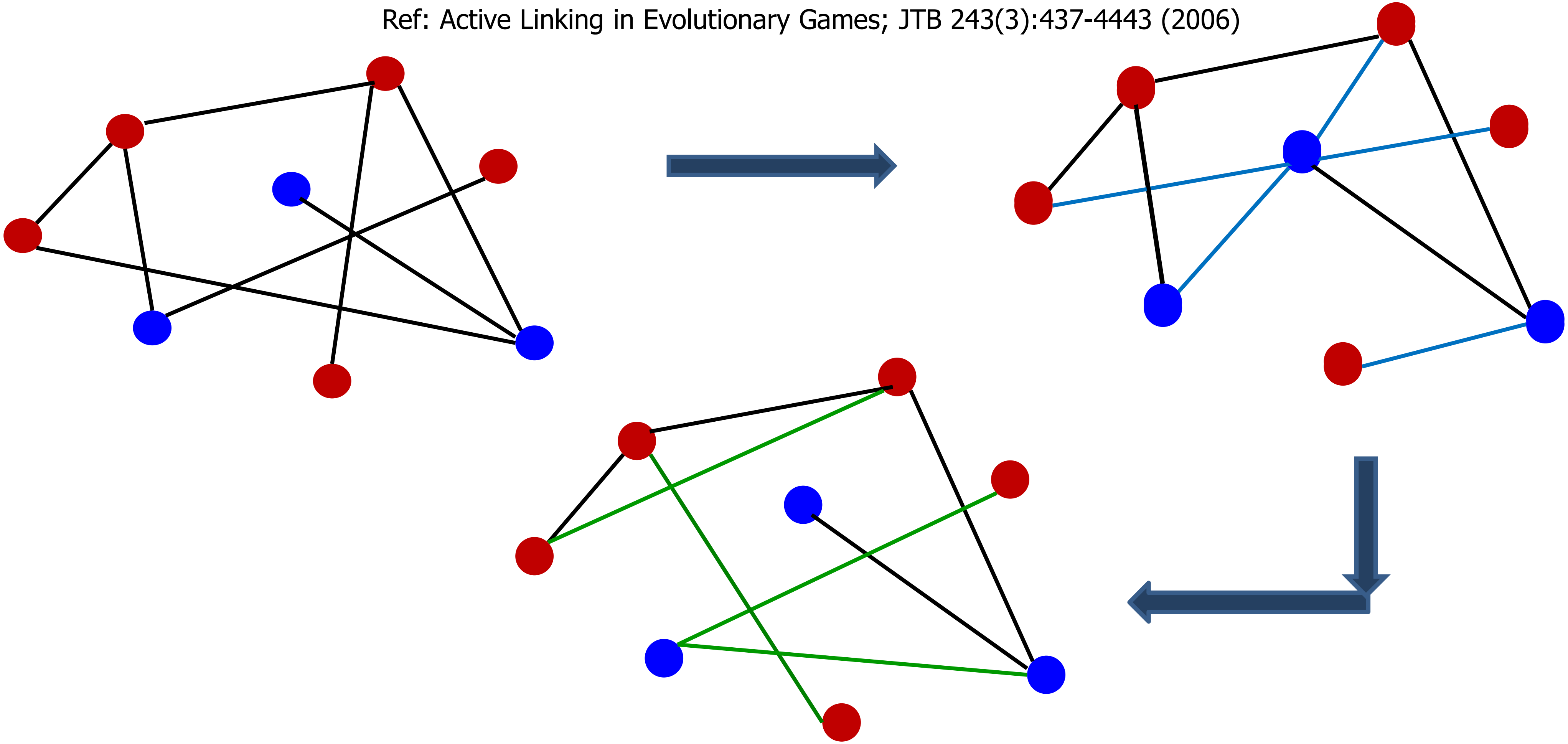


Arrow indicates b/c=k. Theoretical Prediction : b/c > k ➔ $\rho_C > \frac{1}{N} > \rho_D$

- ❖ Discrepancy with theoretical prediction observed for *non-regular* graphs
- ❖ Discrepancy increases with increasing k but decreases with increasing N

Evolutionary Games on Networks with Dynamic Linking

Ref: Active Linking in Evolutionary Games; JTB 243(3):437-4443 (2006)



Two Relevant Time-Scales

Time-Scale of Linking Dynamics: τ_a

Time-Scale of Strategy update Dynamics: τ_e

$$\tau_a >> \tau_e$$

⇓

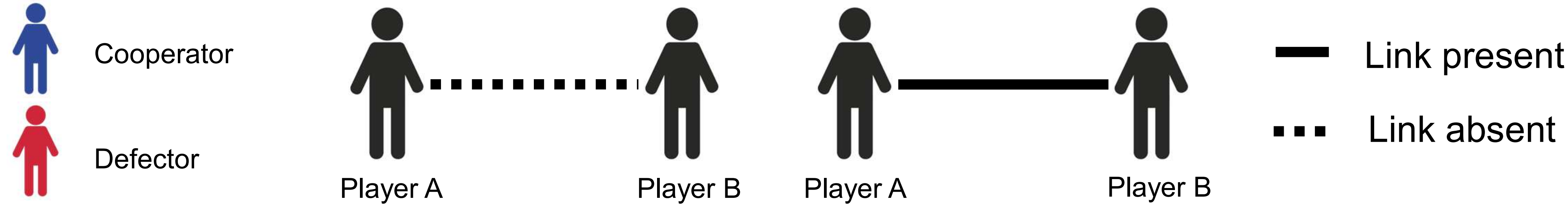
Effectively Static Network

$$\tau_a << \tau_e$$

⇓

Links equilibrate before strategy updates occur

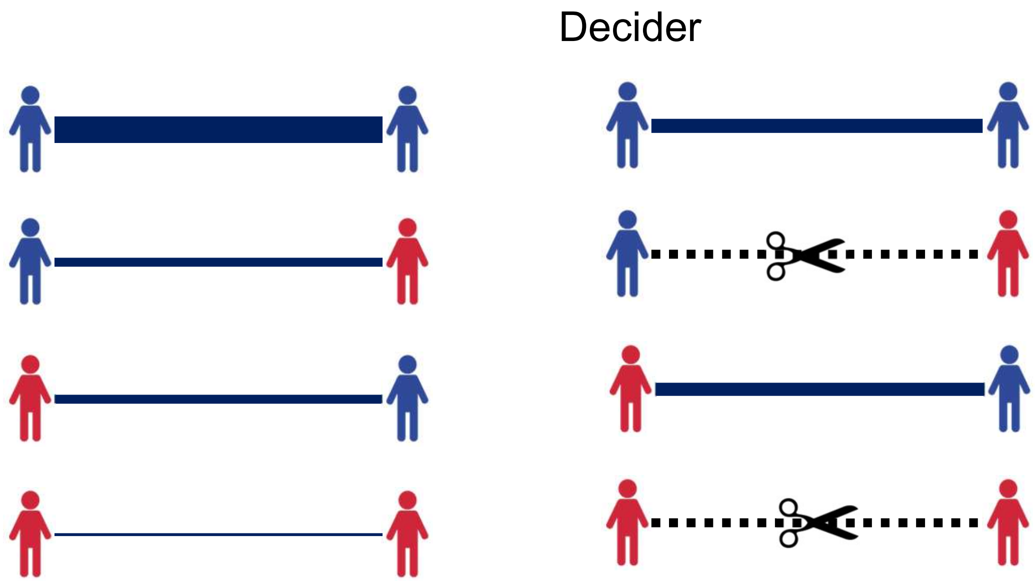
Updating social ties by making and breaking links



$P_{ij}^{\ m}$

Probability of *making* a new ink

$P_{ij}^{\ b}$ related to $\alpha_i \alpha_j$



$P_{ij}^{\ b}$

Probability of *breaking* an existing ink

$P_{ij}^{\ b}$ related to τ_{ij}

$$P_{CC}^{\ m} > P_{CD}^{\ m} = P_{DC}^{\ m} > P_{DD}^{\ m}$$

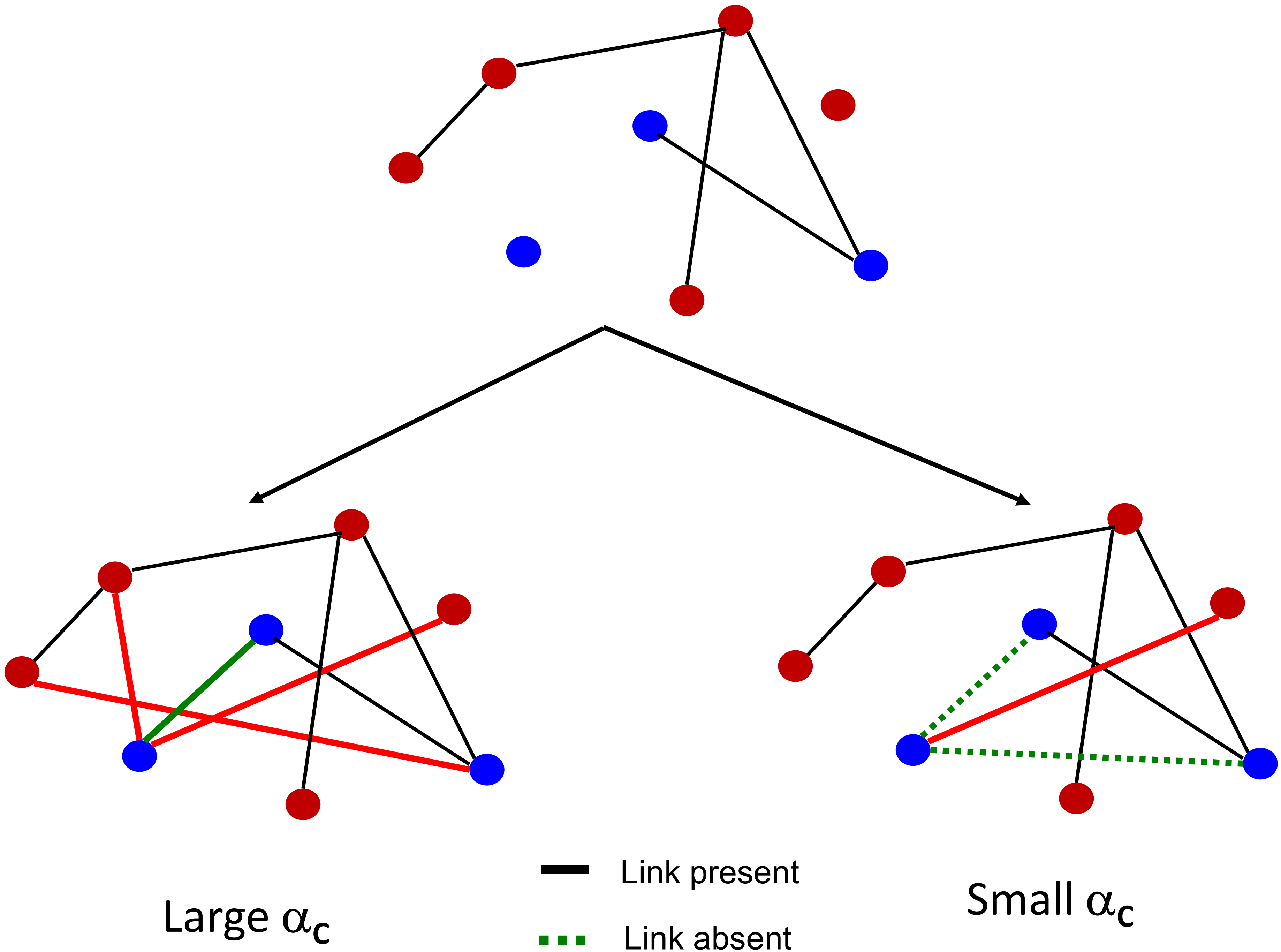
$$\alpha_C^{\ 2} > \alpha_C \alpha_D > \alpha_D^{\ 2}$$

$$P_{CC}^{\ b} < P_{CD}^{\ b}$$

$$P_{DC}^{\ b} < P_{DD}^{\ b}$$

Special case: $\alpha_C = \alpha_D = \alpha$

Updating social ties by making and breaking links



Initialize (I), Interact(I), Update (U), Repeat (R)

IIUR Structure of an agent-based simulation (ABS) of evolutionary games on networks

- Initial configuration: Specify **Initial** distribution of different strategies on a network

- Payoff calculation: **Interaction** and payoff calculation for every agent in the population

- Strategy update: **Update** strategies of agents using specified deterministic or stochastic update rules

synchronous *asynchronous*

Spatial Games

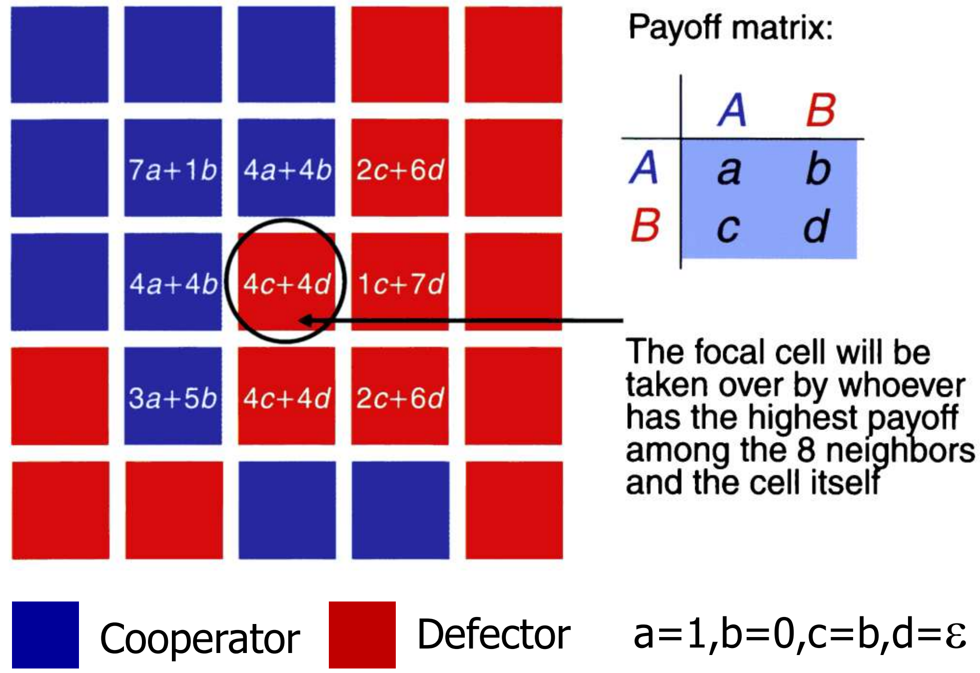
Rules for *Deterministic* Spatial Games

1. The payoff to each player is given by the total payoff obtained by playing each of its eight neighbours.

2. Rules for updating a cell are deterministic: The focal (central) cell is replaced either by itself or one of the eight neighbouring cells (Moore neighbourhood) depending on which has the highest payoff.

3. All cells are updated *simultaneously* (synchronous updating)

4. Periodic boundary condition is used to ensure all cells are treated in the same way and there are no boundary effects.



$a=1, b=0, c=b, d=\varepsilon$

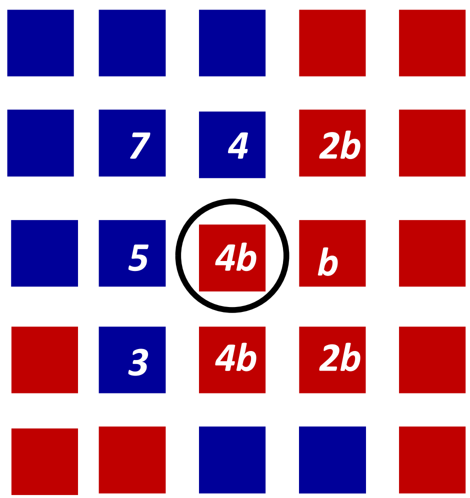
b-Measure of benefit gained from *exploiting* an *altruistic* partner relative to the benefit gained from *cooperating* with an *altruistic* partner

The survival of a cell depends on its own strategy, the strategy of its eight neighbours as well as the strategies of their neighbours ➔ 25 cells in all

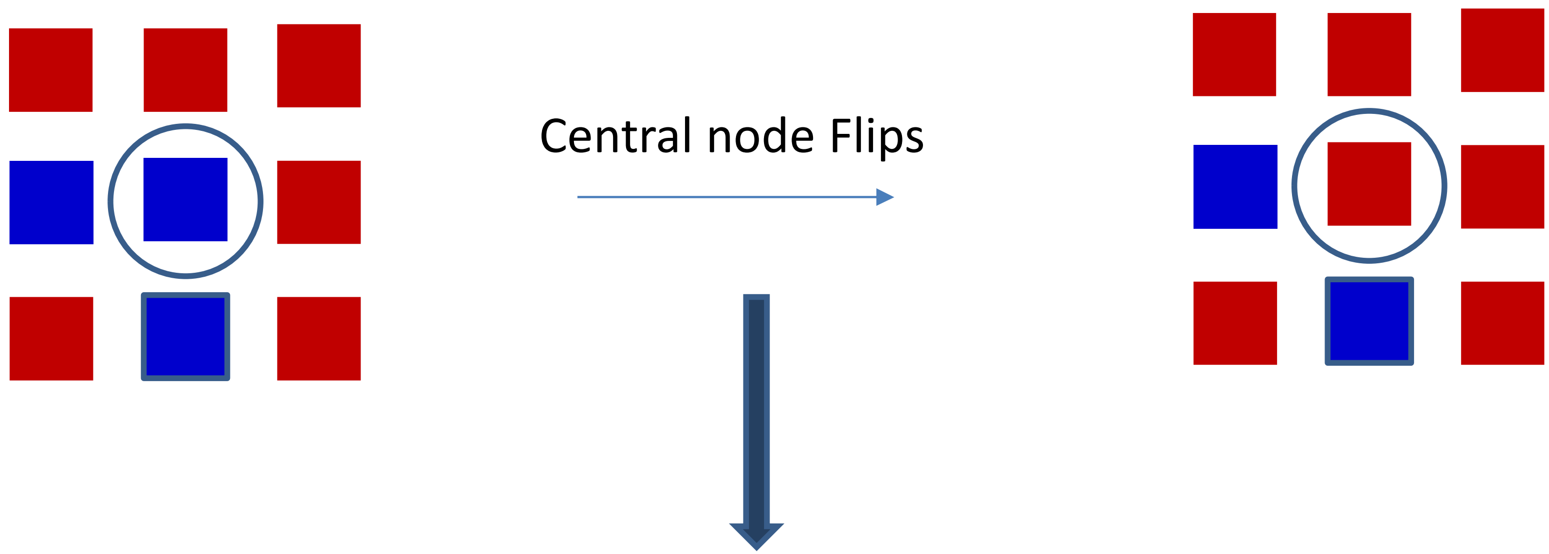
As $\varepsilon \rightarrow 0$, the focal cell (D) has a total payoff = 4b since it is surrounded by 4 C's and 4 D's.

If **4b>7**, central cell remains a Defector in the next generation

If **4b<7**, central cell transforms from Defector to Cooperator in the next generation



Asynchronous update



Neighbourhood changes and payoff changes for all neighbours of the central node

Spatial Games

Rules for *Stochastic* Spatial Games

1. The payoff to each player is given by the total payoff obtained by playing each of its eight neighbours.

2. Rules for updating a cell are stochastic:

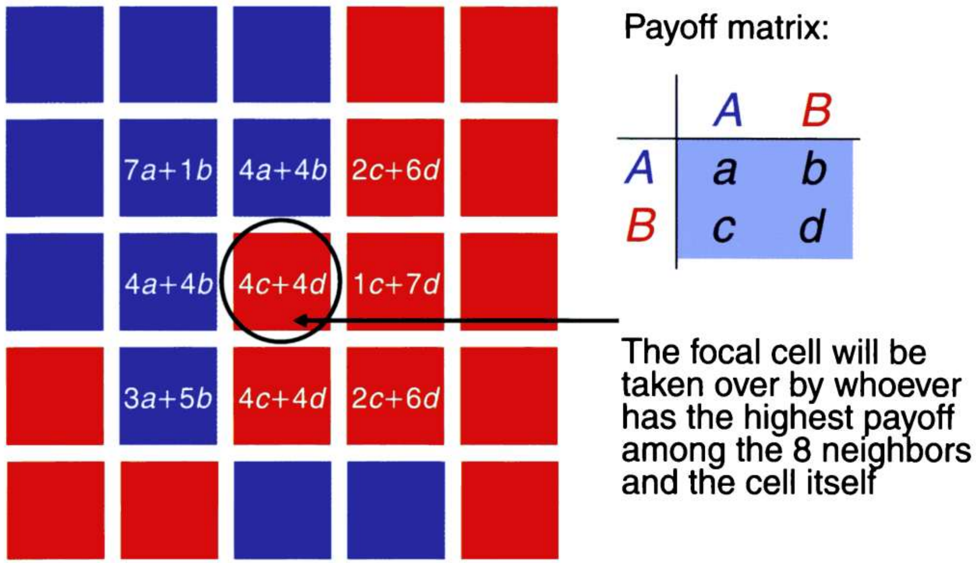
❖ The fitness of all altruists (fC) and all selfish (fD) agents in the neighbourhood of each focal player is separately calculated.

❖ The fractional fitness of the altruists (FC) and selfish agents (FD) is calculated by dividing fC and fD by the total fitness (fC+fD). FC=fC/(fC+fD) etc

❖ Generate random number (RN) between 0 & 1.
IF RN <minimum(FC,FD)
replace focal cell with C if FC<FD
with D if FD<FC
ELSE
replace focal cell with C if FC>FD
with D if FD>FC

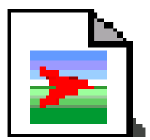
3. All cells are updated *simultaneously* (***synchronous updating***)

4. Periodic boundary condition is used to ensure all cells are treated in the same way and there are no boundary effects.



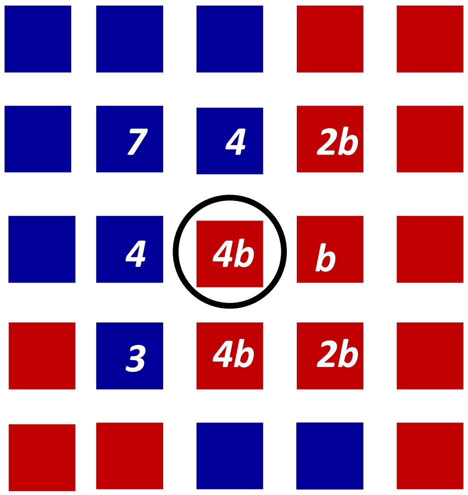
Cooperator Defector a=1,b=0,c=b,d=ε

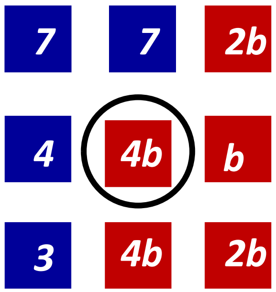
b-Measure of benefit gained from *exploiting* an *altruistic* partner relative to the benefit gained from *cooperating* with an *altruistic* partner



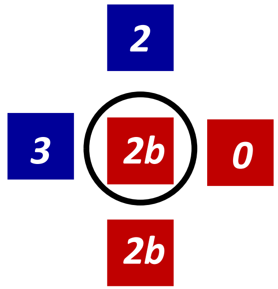
Altruism.nlogo

The Moore and Von-Neumann Neighbourhood





9-cell Moore neighbourhood



5-cell Von-Neumann neighbourhood

Rules for updating a cell: The focal (central) cell is replaced either by itself or one of the ***eight*** neighbouring cells (Moore neighbourhood) depending on which has the highest payoff. The cells diagonally adjacent to the focal cell are ***also*** part of the neighbourhood.

Rules for updating a cell: The focal (central) cell is replaced either by itself or one of the ***four*** neighbouring cells (Von-Neumann neighbourhood) depending on which has the highest payoff. The cells diagonally adjacent to the focal cell are ***not*** part of the neighbourhood.

Initial Condition: Half the cells are randomly chosen to be cooperators and remaining half as defectors.

Colour Code:

Blue : C that was C in the previous generation.

Green: C that was D in the previous generation.

Red: D that was D in the previous generation.

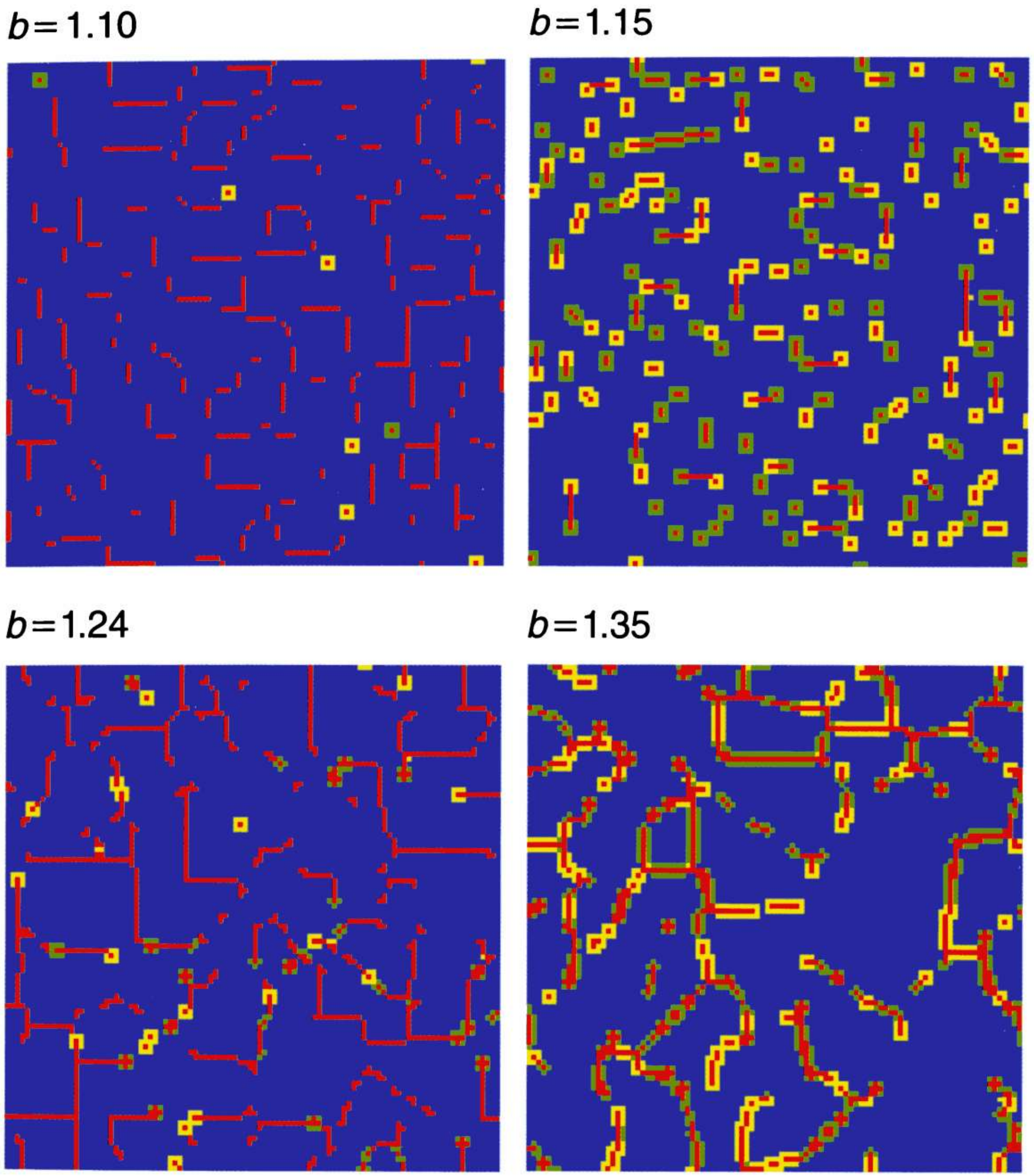
Yellow: D that was C in the previous generation.

b=1.10: Oscillation from isolated single defectors to squares of 9 defector and then back to single defector.

b=1.24: Larger lines of still mostly disconnected defectors are observed.

b=1.35: Lines of defectors now form a network with oscillating islands (yellow and green regions) around lines of defectors.

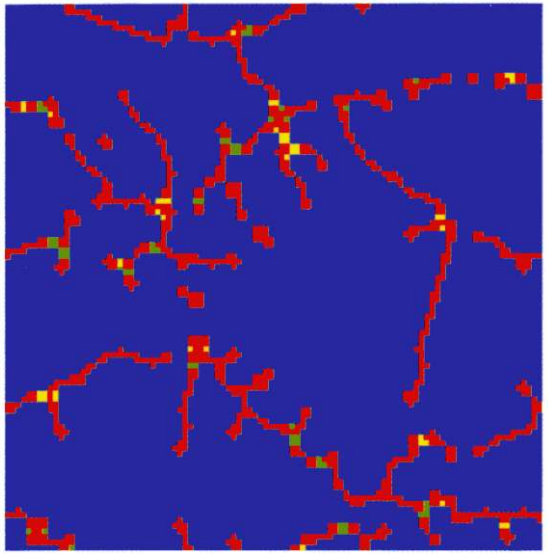
b the only parameter determining the evolution of the spatial distribution of C's and D's



100x100 Lattice

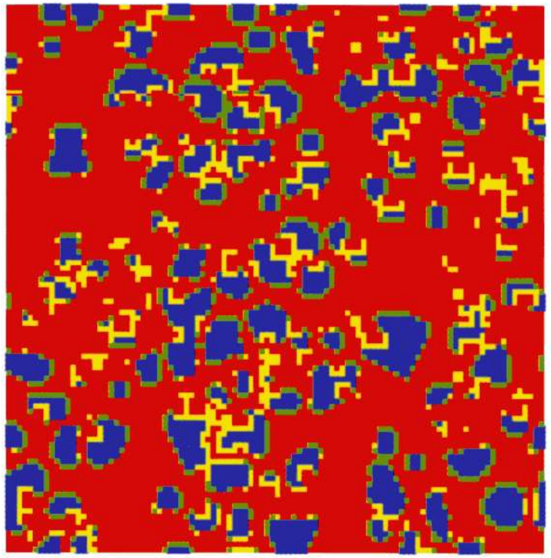
b=1.55: Mostly static network of defectors in large islands of cooperators.

$b = 1.55$



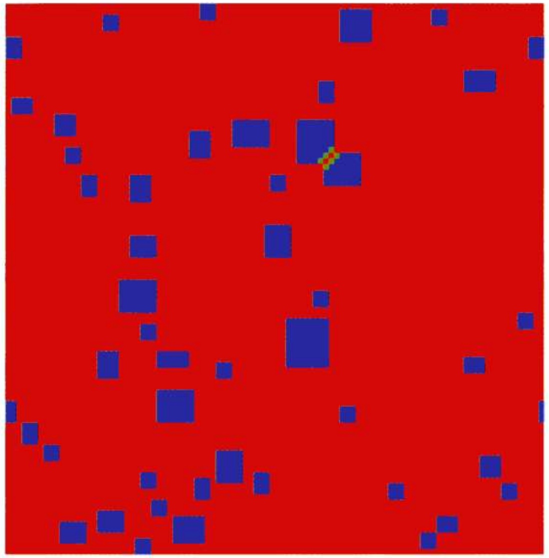
b=1.65: Defectors have attained majority by replacing most of the cooperators. Configuration with dynamic clusters of cooperators seen.

$b = 1.65$



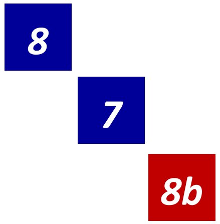
b=1.70: Static pattern showing a few clusters of cooperators in a sea of defectors. Lines of defectors now form a network with oscillating islands (yellow and green regions) around lines of defectors.

$b = 1.70$



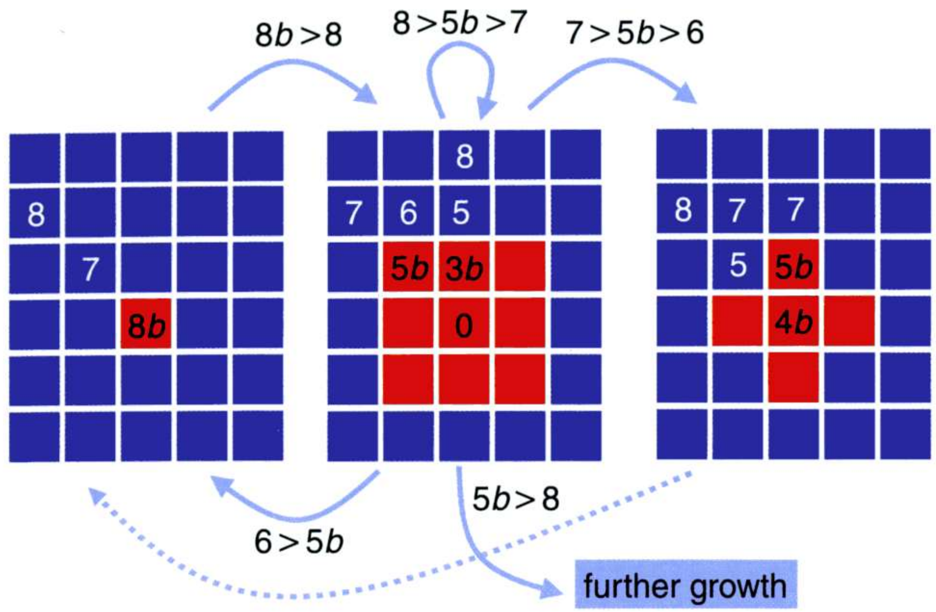
Defectors invading Cooperators

What is the likelihood that a single Defector mutant will take over a population consisting of Cooperators ?



If 7<8b<8, cell in the middle will remain C

If 8b>8, cell in the middle will become D, single D in the centre expands to give a cluster of 9 D's.



9D

5b<6: Red cell with payoff 5b is replaced by blue cell with payoff 6:

9D➔1D

7>5b>6: Red cell at the corners with payoff 5b replaced by blue cell with payoff 7:

9D➔5D

8>5b>7: 9D configuration remains unchanged

9D➔9D

5b>8: Blue cell with payoff 7 replaced by red cell with payoff 5b:

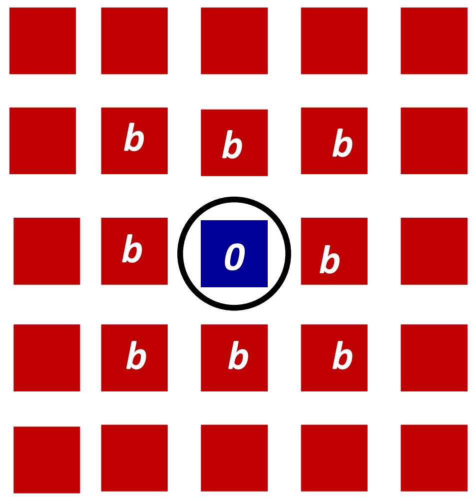
9D➔25D

Cooperators invading Defectors

A single cooperator can never survive in a population of defectors and is immediately eliminated.

Cooperators can only survive in clusters.

Sometimes the cluster of cooperators can grow in size.



9C

2b<3: Red cells with payoff 2b,3b are replaced by blue cells with payoff 3,5:

9C➔25C

3/2<b<5/3: Red cell with payoff 3b replaced by blue cell with payoff 5: **C** cluster expands along central axis but not along diagonals

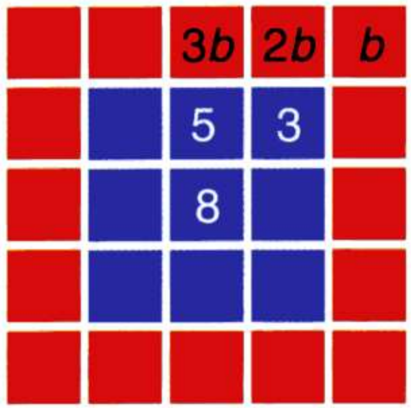
9C➔21C

5/3<b<8/3: 9C configuration remains unchanged

9C➔9C

b>8/3: 9C cluster disappears

9C➔0C



b-dependence on Spatial Evolution of Cooperators and Defectors

(i) b<8/5: Only C clusters keep growing; Cooperators dominate

(ii) b>5/3: Only D clusters keep growing; Defectors dominate

(iii) 8/5<b< 5/3: Both C and D clusters keep growing; *Co-existence* between cooperators and defectors

In (i) and (ii), final frequency of cooperators and defectors depends on the initial configuration.

In (iii) final frequency is independent of initial configuration. The pattern is dynamic but the final frequency remains nearly constant at 30% cooperators ➔ Dynamic Equilibrium

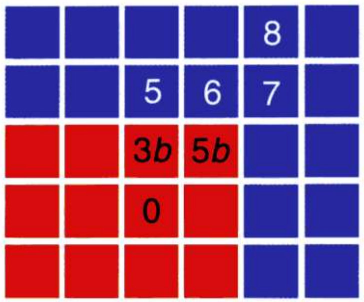
Dynamical Fractals and Evolutionary Kaleidoscopes

For **8/5<b<5/3**, configuration starting from a single defector shows repetitive (fractal-like) patterns.

A single D grows to form a 3x3 square.

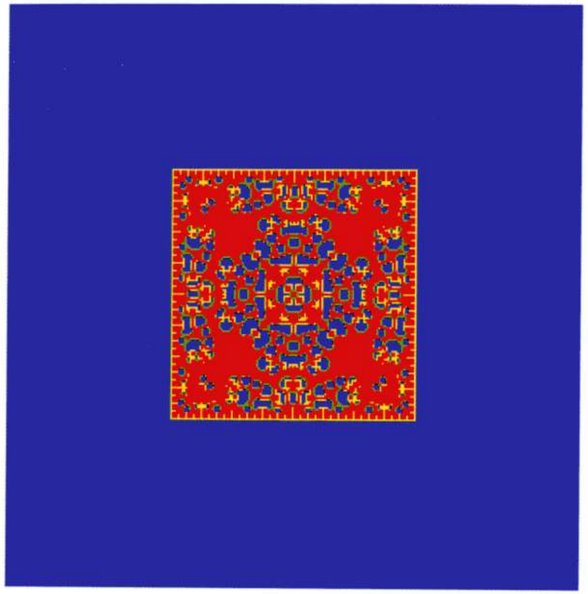
If **5b>8**, D wins at the corners and red cell with payoff 5b replaces the blue corner cell with payoff 7

If **3b<5**, C wins along lines and red cell with payoff 5b is replaced by blue cell with payoff 5

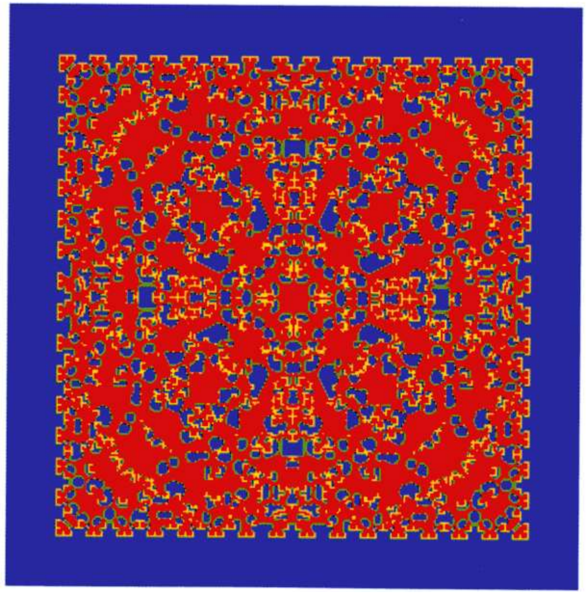


If **8/5<b<5/3 ➔ 1.6<b<1.67,** C wins along lines but loose along irregular boundaries.

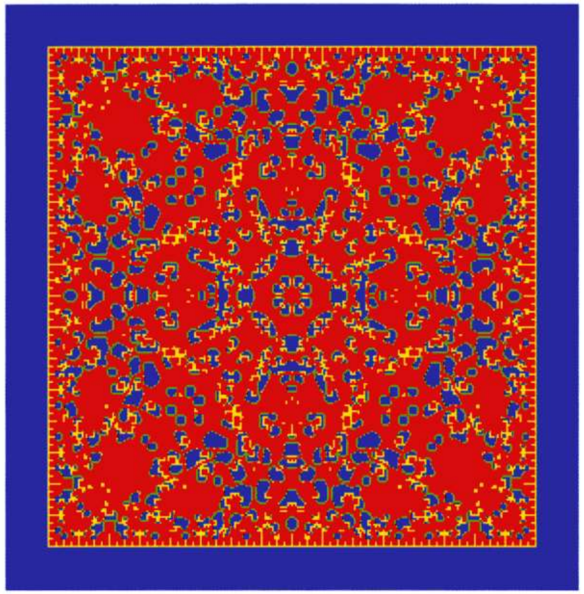
$t = 64$



$t = 124$



$t = 128$

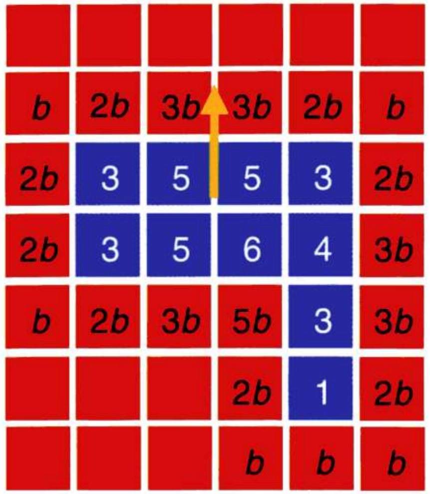


Invasion by Cooperators

Two walkers can collide and generate two large clusters of cooperators.

A walker is a cluster of 10 cooperators moving in a direction shown by the arrow.

A “walker”



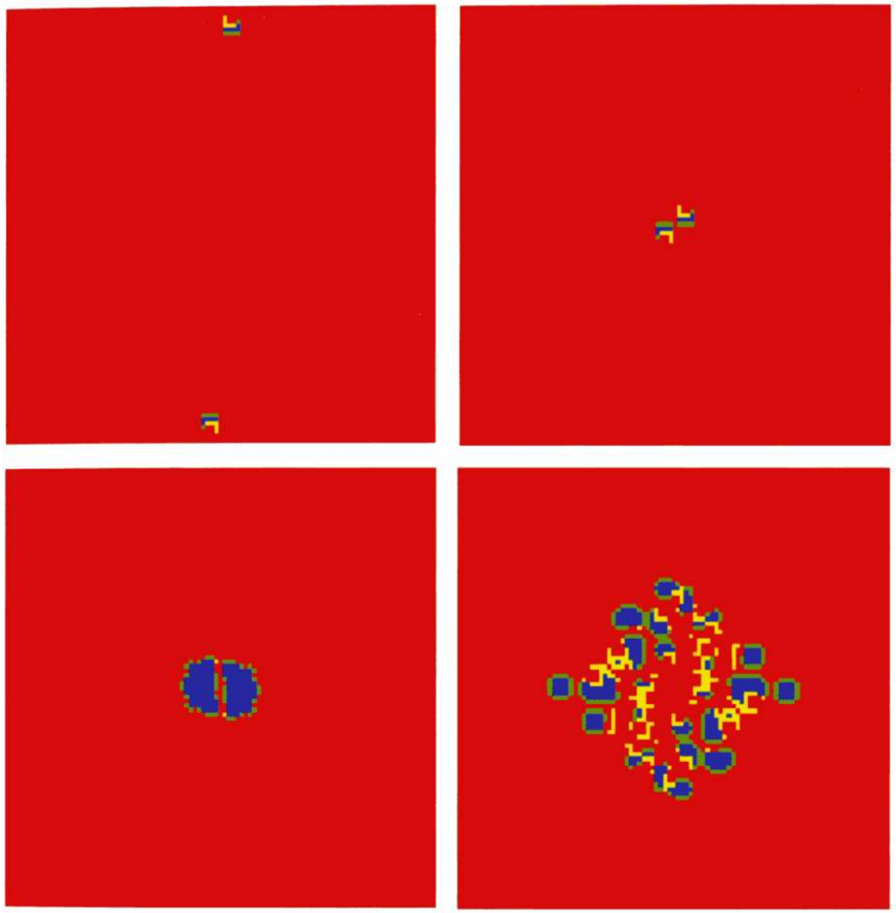
$8/5 < b < 5/3$

$3b<5$ since $b<5/3$

➔ $2b<5$

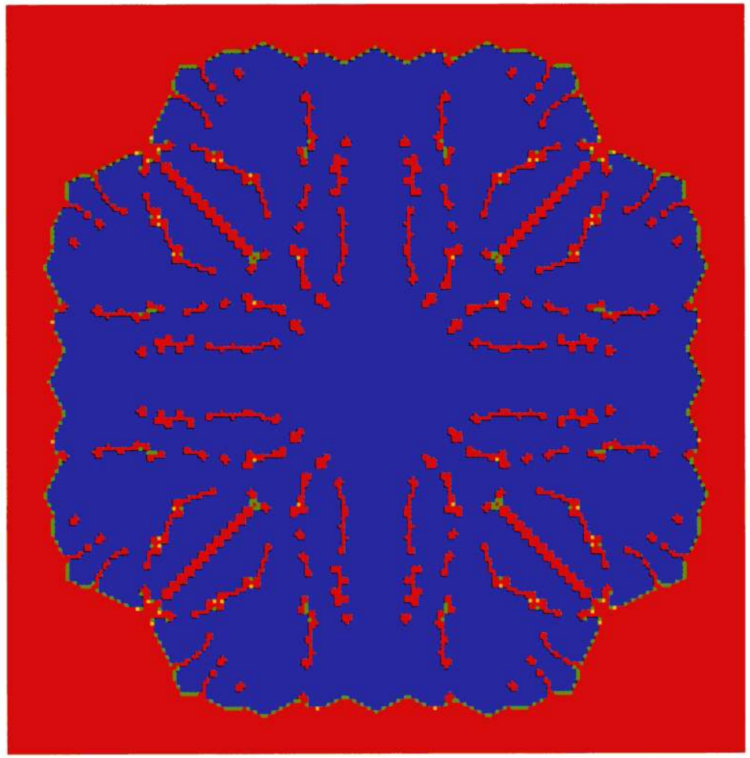
$2b>3$ since $b>1.6$ implies **C**[payoff=1]➔**D**

A collision of two “walkers” can lead to a “big bang” of cooperation



Parameter region: **$3/2<b<8/5$**

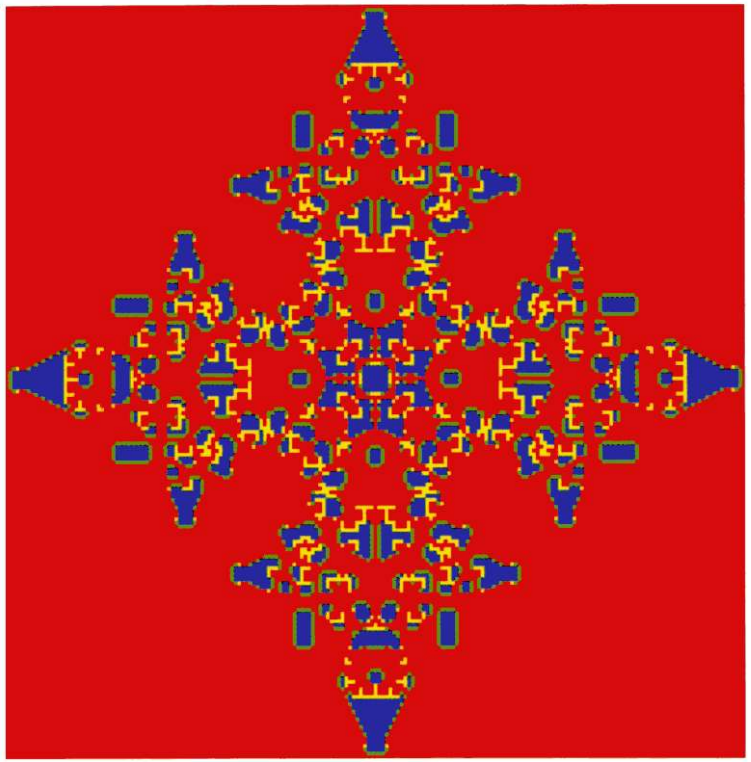
Initial Configuration: Single cluster of 3x3 C's



Increasing ***b***

Parameter region: **$8/5<b<5/3$**

Initial Configuration: Single cluster of 3x3 C's



Summary

A structured population can facilitate survival and sustenance of cooperators even under conditions that are unfavourable for their survival in a mixed-population scenario

Cooperators survive by forming clusters (➔ strength in numbers) and the growth or shrinkage of the clusters depend on interactions at the boundary of the cluster

The extent of survival and spread of cooperators depend on the relative advantage that a selfish agent has over a cooperator/altruist

The greater the advantage, the lesser is the likelihood of survival of cooperators even in structured populations

Connection between ABS models and Stochastic Reaction-Diffusion Models

Fully Stochastic Agent-Based Simulation Models

Sometimes an *exact* correspondence exists

Stochastic Reaction-Diffusion (SRDE) Models

Exact correspondence

Results of an SRDE model are completely consistent with that of the ABS model for moderate to large population size (N)

Correspondence can be established for a Rock-Paper-Scissors Game

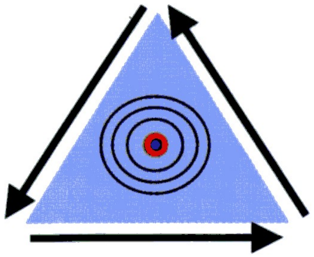
A useful framework for understanding the conditions for multi-species coexistence in an ecosystem

Rock Paper Scissors: 3x3 Game with Cyclic Domination

Example of an Evolutionary Game with non-transitive interactions

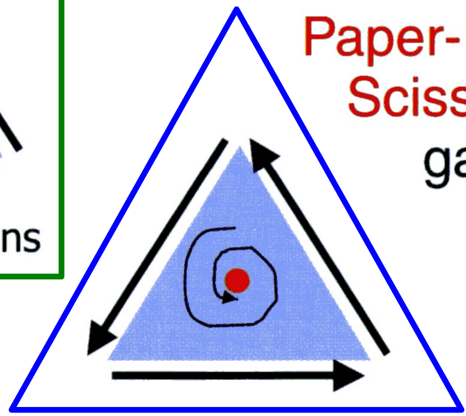
$$A = \begin{array}{c} \\ R \\ S \\ P \end{array} \begin{array}{c} \begin{array}{ccc} R & S & P \end{array} \\ \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \end{array}$$

Det[A]=0 ➔ $a_1 a_2 a_3 = b_1 b_2 b_3$



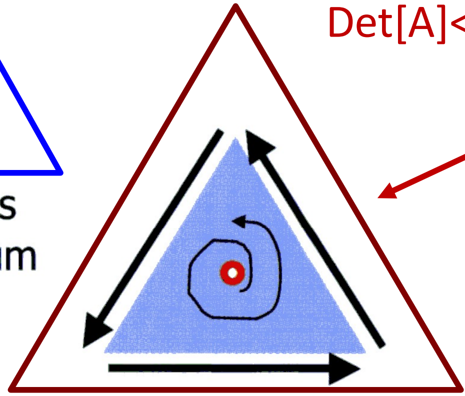
Neutral oscillations

The Rock-Paper-Scissors game



Damped oscillations to a stable equilibrium

Det[A]<0 ➔ $a_1 a_2 a_3 > b_1 b_2 b_3$



Increasing oscillations to a heteroclinic cycle

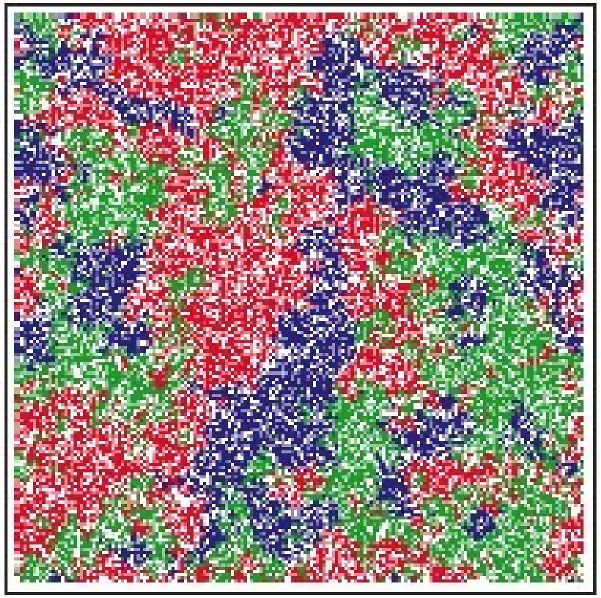
$$A = \begin{pmatrix} 0 & -a_2 & b_3 \\ b_1 & 0 & -a_3 \\ -a_1 & b_2 & 0 \end{pmatrix}$$

Det[A]>0 ➔ $a_1 a_2 a_3 < b_1 b_2 b_3$

Local dispersal promotes biodiversity in a real-life game of rock–paper–scissors

Benjamin Kerr*, Margaret A. Riley†, Marcus W. Feldman* & Brendan J. M. Bohannan*

A bactericidal model



C - colicin producing *E.coli*

S - colicin sensitive *E.coli*

R - colicin resistant *E. coli*

Update Dynamics

Focal lattice site (F) chosen **at random**

If F is **empty:**

1. **Local** update: It is filled with a cell of type i=C or S or R from its ***local* neighbourhood** (8 nearest neighbours surrounding F) with a **probability proportional to fraction (f_i) of cell-type (i) present in the neighbourhood**.

3. **Global update:** It is filled with a cell of type i=**C** or **S** or **R** from its ***global* neighbourhood** (anywhere in the lattice except the focal site) with a **probability proportional to fraction (f_i) of cell-type (i) present in the neighbourhood.**

Generate random number (RN) between [0,1]

- **If RN<f_C/(f_C+ f_S+ f_R) ➔ empty cell is filled with C**
- **If f_C/(f_C+ f_S+ f_R) < RN ≤ (f_C+f_S)/(f_C+ f_S+ f_R)**

➔ empty cell is filled with S

- **If (f_C+f_S)/(f_C+ f_S+ f_R) <RN ≤1 ➔ empty cell is filled with R**

If focal lattice site F is **occupied** by cell-type i,

it is killed with probability Δ_i

$$\Delta_C = 0.33 \qquad \Delta_R = 0.3125 \qquad \Delta_S = \Delta_{S0} + \tau f_C ; \Delta_{S0} = 0.25$$

$\tau =$ Measure of toxicity of colicin

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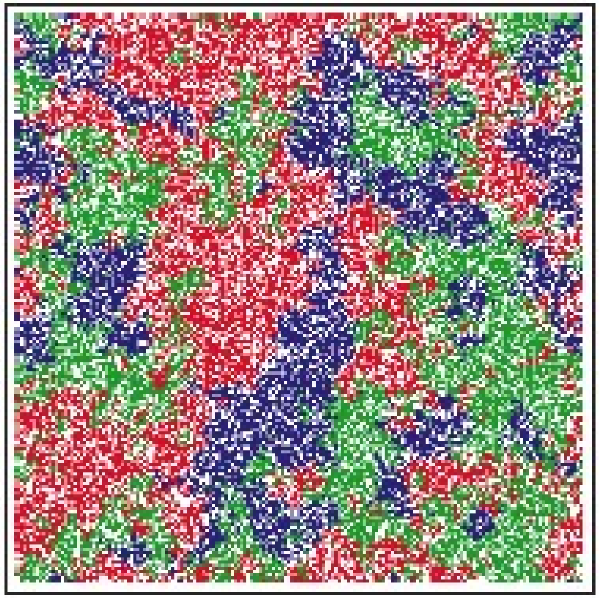

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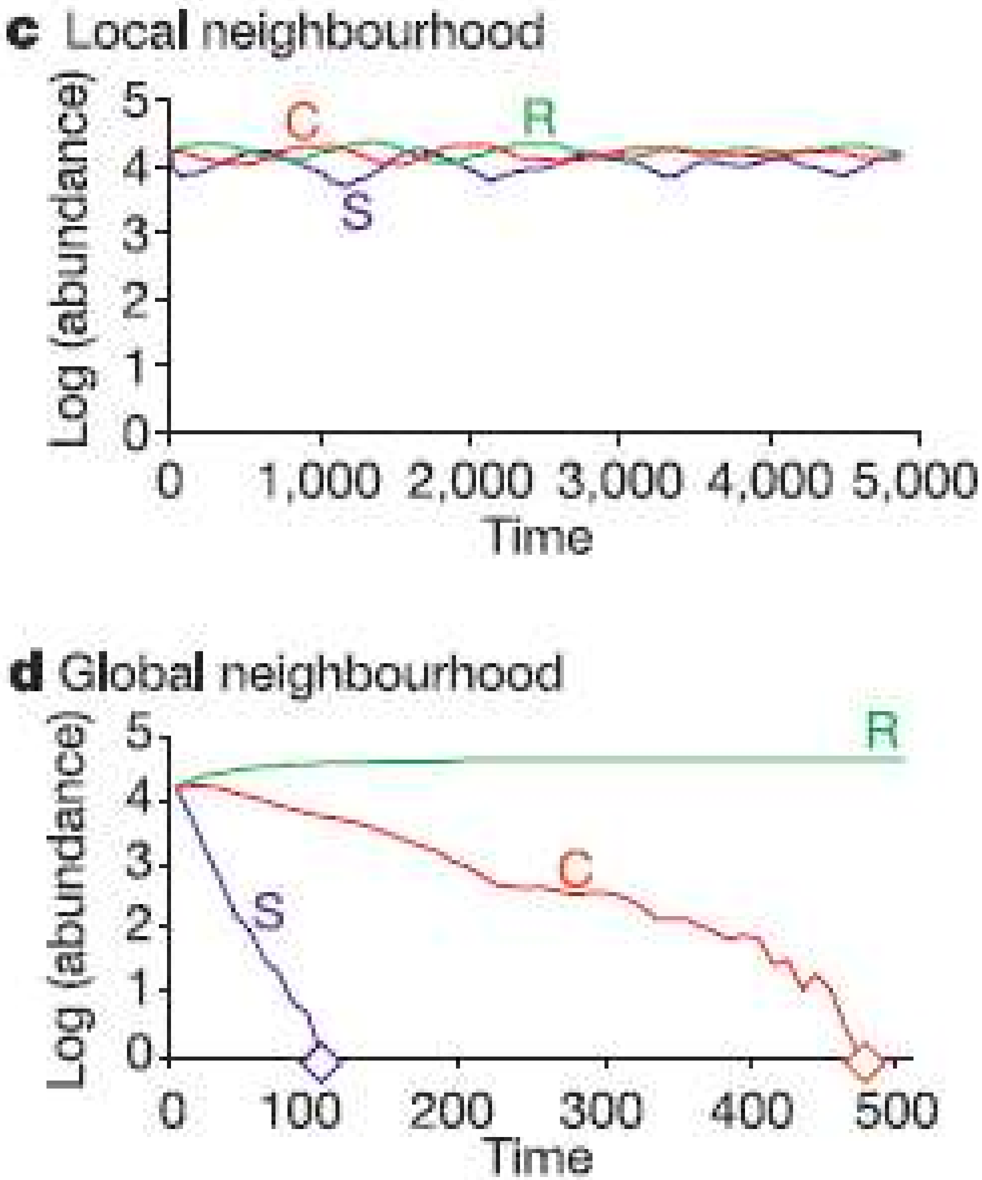
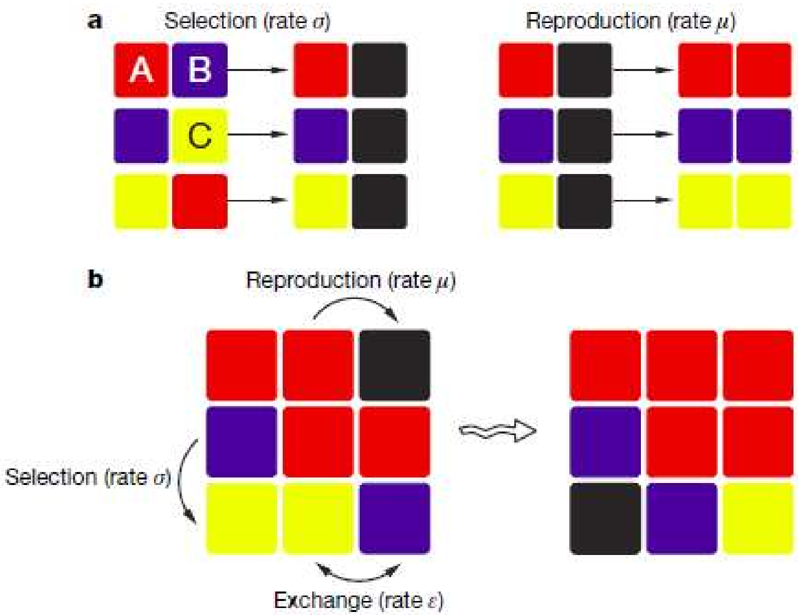


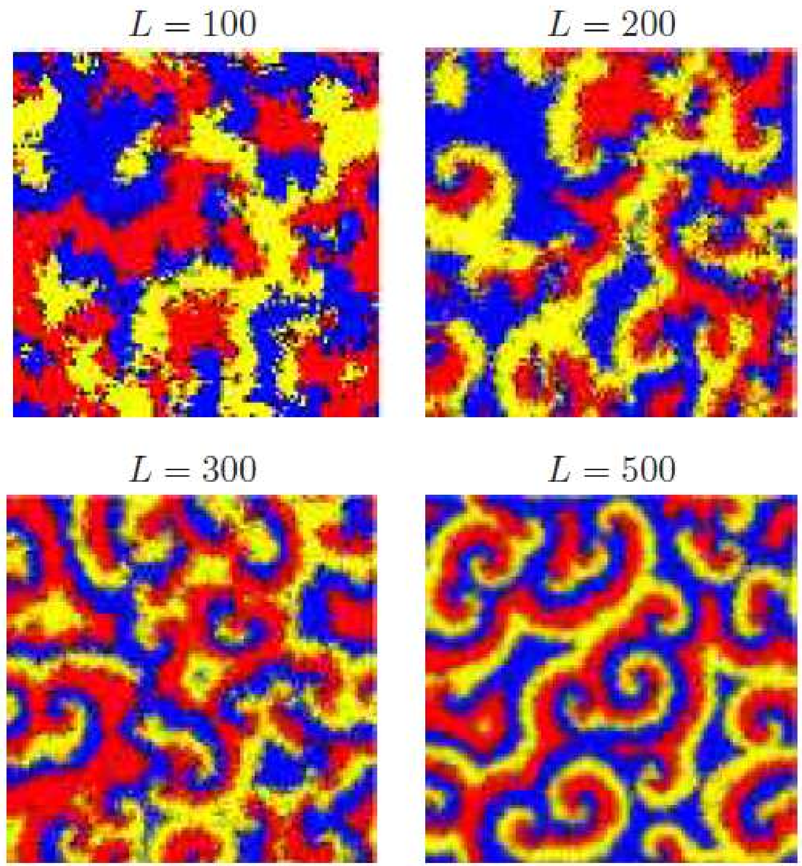
Rock Paper Scissors on a Spatial 2D Lattice with Diffusion



May-Leonard Model

$$AB \xrightarrow{\sigma} A\oslash\,,\qquad A\oslash \xrightarrow{\mu} AA$$
$$BC \xrightarrow{\sigma} B\oslash\,,\qquad B\oslash \xrightarrow{\mu} BB$$
$$CA \xrightarrow{\sigma} C\oslash\,,\qquad C\oslash \xrightarrow{\mu} CC$$

Selection   Reproduction



Effect of Lattice Size on Pattern Formation

Shows important role of noise in perturbing the patterns for small values of L

Update Rules:
Choose one of 3 reactions with the following probabilities

Reproduction

$$\frac{\mu}{\mu+\sigma+\varepsilon}$$

Selection

$$\frac{\sigma}{\mu+\sigma+\varepsilon}$$

Exchange

$$\frac{\varepsilon}{\mu+\sigma+\varepsilon}$$

Reichenbach, Mobilia, Frey; Nature 2007, Journal of Theoretical Biology 2008

Alternative Model

$$AB \xrightarrow{p} A\emptyset\,,\quad BC \xrightarrow{p} B\emptyset\,,\quad CA \xrightarrow{p} C\emptyset,$$
$$AB \xrightarrow{z} AA\,,\quad BC \xrightarrow{z} BB\,,\quad CA \xrightarrow{z} CC,$$
$$A\emptyset \xrightarrow{q} AA\,,\quad B\emptyset \xrightarrow{q} BB\,,\quad C\emptyset \xrightarrow{q} CC,$$

$$A \xrightarrow{\mu} \begin{cases} B \\ C \end{cases},\quad B \xrightarrow{\mu} \begin{cases} A \\ C \end{cases},\quad C \xrightarrow{\mu} \begin{cases} A \\ B \end{cases}$$

Relation between Deterministic and Stochastic Spatial Models of Rock-Paper-Scissors Game

Deterministic

↓ Noise incorporated through a Gaussian noise term in the evolution equations

Partially Stochastic represented by Stochastic PDE's

↓

Fully Stochastic Agent-Based Simulations (ABS)

Rules for interaction and update are explicitly specified for each agent

Relation between the Master Equation and SPDE's

Master Equation

Kramers-Moyal expansion of Master Equation to second order in (δs) i.e. O(1/N)

Fokker-Planck (FP) Equation

Stochastic Differential Equations SDE's

Incorporate diffusion through exchange process to leading order in ($\delta s \equiv 1/N$) i.e. $(1/N)^0$

Partially Stochastic Model represented by Stochastic PDE's

➔ Noise due to number fluctuations incorporated to $O(\frac{1}{\sqrt{N}})$

Model Assumptions

- Each lattice site can be occupied by at most 1 A or 1B or 1 C
- Every agent at an occupied site reacts once per time step
- Any one of 4 possible reactions can occur at a filled site
  - Death occurring with probability λ
  - Reproduction occurring with a probability μ/(μ+σ+ε)
  - Selection occurring with a probability σ/(μ+σ+ε)
  - Exchange occurring with a probability ε/(μ+σ+ε)

Objective: Derive the following set of SPDE's from first principles

$$\partial_t a = -D\nabla^2 a + \alpha_A(s) + C_{AA}\xi_A \qquad \xi_A(\vec{x},t)\xi_A(x',t') = \delta(x-x')\delta(t-t')$$

$$\partial_t b = -D\nabla^2 b + \alpha_B(s) + C_{BB}\xi_B$$

$$\partial_t c = -D\nabla^2 c + \alpha_C(s) + C_{CC}\xi_C$$

D, $\alpha_i(s)$, $C_{AA}(s)$ to be determined in terms of a,b,c and model parameters

Fixing the length scale

L= # lattice sites along each spatial dimension, N = Total #lattice points

d= #spatial dimensions, δr = lattice spacing;

$N=L^d$

If the length of each side is set to 1 i.e. $L\,\delta r = 1$ ➔ fixes the length scale

$L\,\delta r = 1 \Rightarrow N^{1/d}\,\delta r = 1 \Rightarrow \delta r = N^{-(1/d)}$

For d=2: $\delta r = \frac{1}{\sqrt{N}} \Rightarrow \delta r^2 = N^{-1}$

Source of Noise in SPDE's

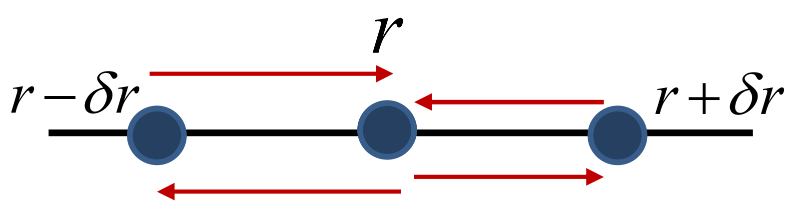
❖ Noise due to number fluctuations arising from birth-death processes

Appears in SDE at $O(\frac{1}{\sqrt{N}})$

❖ Noise due to number fluctuations arising from exchange processes

The Diffusion term to leading order i.e. in the absence of fluctuations i.e. $O\left(1/N\right)^0$

$$T_1 = \frac{\varepsilon}{2d}\sum_{i=1}^{d}\left[\left\{a(r+\delta r)-a(r)\right\}+\left\{a(r-\delta r)-a(r)\right\}\right]$$



$$T_1 = \frac{\varepsilon(\delta r)^2}{2d}\sum_{i=1}^{d}\partial_i^2 a(\vec{r},t) \;\; ➔ \;\; T_1 = D\nabla^2 a(\vec{r},t) \quad \text{where} \quad D = \frac{\varepsilon(\delta r)^2}{2d}$$

In the *continuum* limit: $\delta r \to 0$ or equivalently N➔∞; ε ➔ ∞

Exchange processes occur at a much faster time-scale than birth-death processes

➔ exchange processes can be treated deterministically

➔ fluctuations due to exchange processes that arising at $O\left(\frac{1}{N}\right)$ can be ignored

Noise due to number fluctuations arising from exchange processes comes from the term

$$B(\vec{r},\vec{r}') = \frac{D}{N^2}\partial_r \partial_{r'}\left[\delta(\vec{r}-\vec{r}')a(\vec{r})(1-a(\vec{r}))\right]$$

Since noise is proportional to $\sqrt{B(\vec{r},\vec{r}')}$ it appears at $O(\frac{1}{N})$ and can be ignored in comparison to noise due to birth-death processes

Fokker-Planck Equation from the Master Equation

Master Equation

$$\partial_t P(s,t) = \sum \{ P(s+\delta s,t) W(s+\delta s \to s) - P(s,t) W(s \to s+\delta s) \}$$

$$W(s+\delta s \to s) = NT^{+}(s) \qquad W(s \to s+\delta s) = NT^{-}(s)$$

Fokker-Planck Equation

$$\partial_t P(s,t) = -\partial_i \left[\alpha_i(s) P(s,t) \right] + \frac{1}{2} \partial_i \partial_j \left[\beta_{ij}(s) P(s,t) \right]$$

$$\alpha_i(s) = T_i^{+} - T_i^{-} \qquad \beta_{ii}(s) = (T_i^{+} + T_i^{-}) / N$$

Incorporate diffusion through exchange process to leading order in ($\delta s \equiv 1/N$) i.e. $O(1/\mathrm{N})^0$

Langevin Equations (SPDE)

$$\partial_t a = -D\nabla^2 a + \alpha_A(s) + C_{AA}\xi_A$$

$$\partial_t b = -D\nabla^2 b + \alpha_B(s) + C_{BB}\xi_B$$

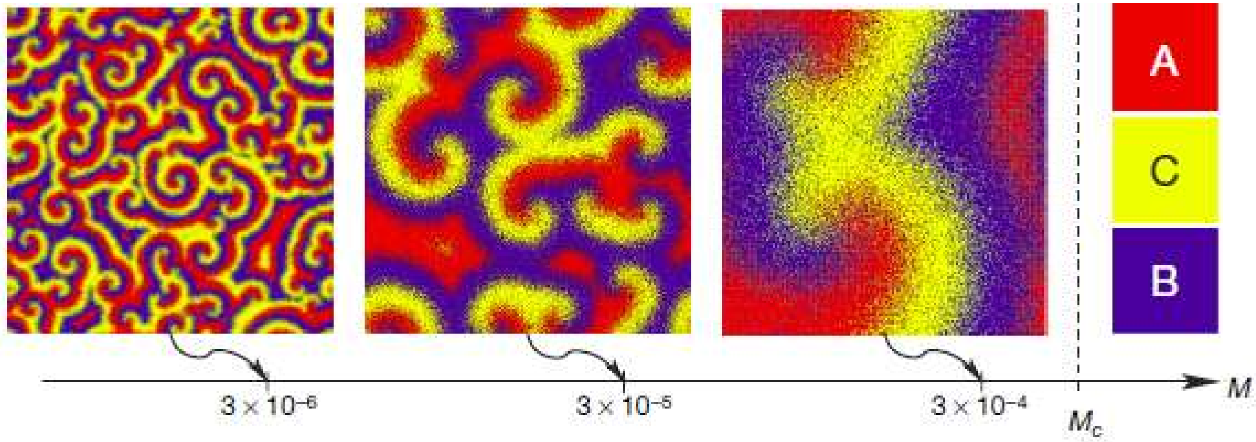
$$\partial_t c = -D\nabla^2 c + \alpha_C(s) + C_{CC}\xi_C$$

$$\xi_A(\vec{x},t)\xi_A(x',t') = \delta(x-x')\delta(t-t')$$

➔ Noise due to number fluctuations incorporated to $\mathrm{O}(\frac{1}{\sqrt{N}})$

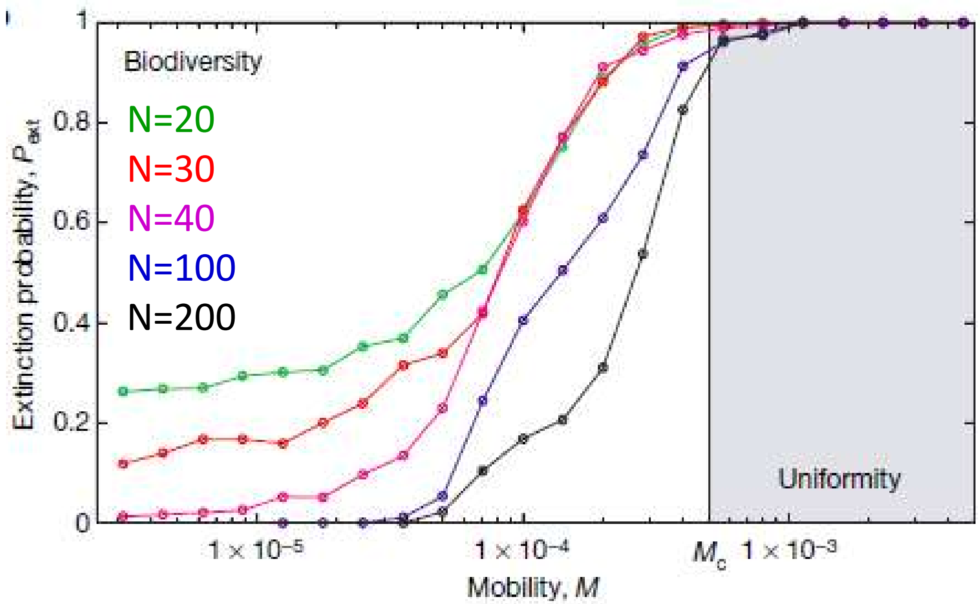
$$C_{ii}(s) = \sqrt{\beta_{ii}}$$

Effect of Increasing Diffusion Coefficient: Loss of Diversity



$M = 2 \in N^{-1} = 4D$: Mean-square dist. in 2d = Area covered by random walker in unit time in 2d

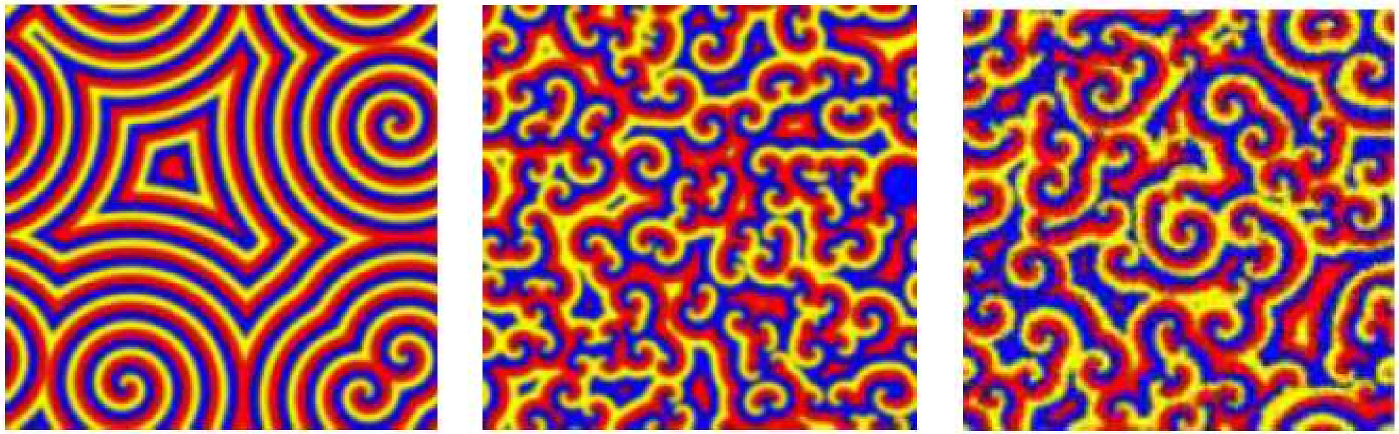
D **:** Diffusion Coefficient

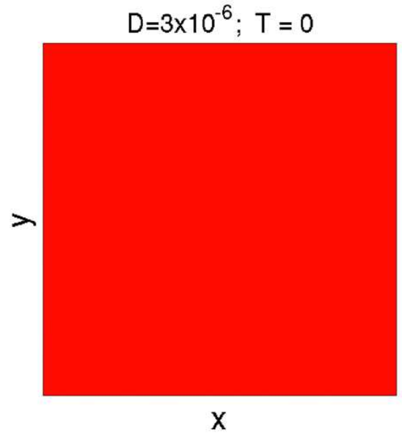


Reichenbach, Mobilia, Frey; Nature 2007, Journal of Theoretical Biology 2008

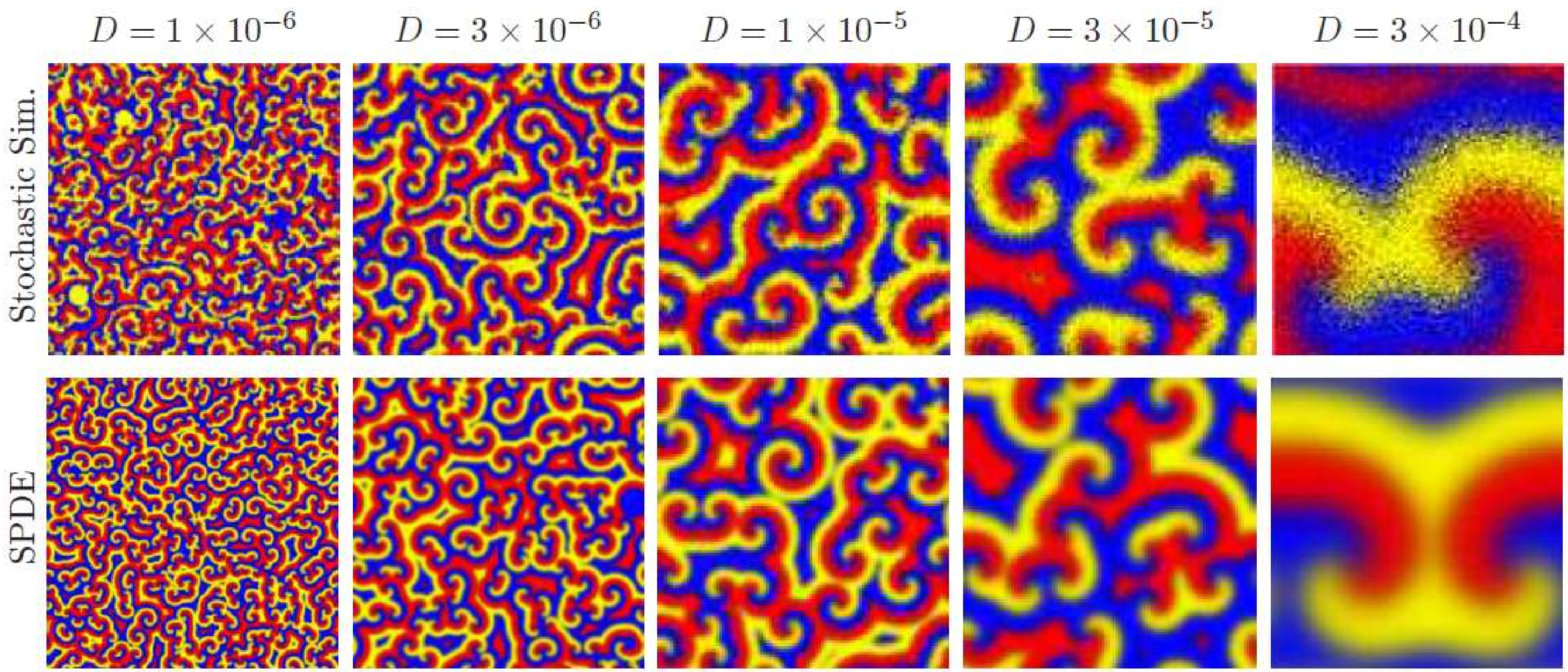
Pattern Formation *without* and *with* Noise

PDE SPDE Stochastic Simulations





$D = 3 \times 10^{-6}$ $\sigma = \mu = 1$



L=1000 L=1000 L=500 L=500 L=300

Solving SPDE's with XMDS2: Code for RPS model of Reichenbach et al 2007

http://www.xmds.org/index.html

```
<?xml version="1.0" encoding="UTF-8"?>
<simulation xmds-version="2">
  <name>RPS_spde_delT</name>

  <author>Prateek Verma</author>
  <description>
    Rock-Paper-Scissor dynamics using PDE.
  </description>

  <features>
      <benchmark />
      <bing />
      <fftw plan="patient" />
      <openmp />
      <auto_vectorise />
      <globals>
          <![CDATA[
          const double sig = 1.0;
          const double mu = 1.0;
          const double D = 3e-5;
                  const double N = 16384.0; // 16384; // 65536.0;
                  const double delx = 1.0/128.0;
          const double delt = 2000.0/20000.0;
          ]]>
       </globals>
     </features>
```

```
<geometry>
   <propagation_dimension> t </propagation_dimension>
   <transverse_dimensions>
      <dimension name="x" lattice="128"  domain="(0, 1)" />
      <dimension name="y" lattice="128"  domain="(0, 1)" />
   </transverse_dimensions>
 </geometry>

<vector name="concentration" type="complex" dimensions="x y">
  <components> a b c </components>
  <initialisation>
    <![CDATA[
    a = 0.25; //+ (cos(2.0*3.14*x*y))/100.0;
    b = 0.25;
    c = 0.25;
    ]]>
  </initialisation>
</vector>

<noise_vector name="drivingNoise1" dimensions="x y" kind="wiener" type="real"
method="dsfmt" seed="314 159 276">
  <components>Eta1</components>
</noise_vector>

<noise_vector name="drivingNoise2" dimensions="x y" kind="wiener" type="real"
method="dsfmt" seed="315 160 277">
  <components>Eta2</components>
</noise_vector>

<noise_vector name="drivingNoise3" dimensions="x y" kind="wiener" type="real"
method="dsfmt" seed="316 161 279">
  <components>Eta3</components>
</noise_vector>
```

```
<sequence>
<integrate algorithm="SI" iterations="5" interval="2000" steps="20000">
  <samples>5</samples>
  <operators>
    <operator kind="ex">
      <operator_names>Lxx Lyy</operator_names>
      <![CDATA[
        Lxx = -1*D*kx*kx;
              Lyy = -1*D*ky*ky;
      ]]>
  </operator>

    <integration_vectors>concentration</integration_vectors>
        <dependencies>drivingNoise1 drivingNoise2 drivingNoise3</dependencies>

    <![CDATA[
    da_dt = Lxx[a] + Lyy[a] + mu*a*(1-a-b-c) - sig*a*c +
Eta1*delt*delx*sqrt(mu*a*(1-a-b-c) + sig*a*c)/sqrt(N);
        db_dt = Lxx[b] + Lyy[b] + mu*b*(1-a-b-c) - sig*b*a +
Eta2*delt*delx*sqrt(mu*b*(1-a-b-c) + sig*b*a)/sqrt(N);
        dc_dt = Lxx[c] + Lyy[c] + mu*c*(1-a-b-c) - sig*c*b +
Eta3*delt*delx*sqrt(mu*c*(1-a-b-c) + sig*c*b)/sqrt(N);
    ]]>
  </operators>
</integrate>
</sequence>
```

Since $k_x = -i\dfrac{\partial}{\partial x}$

$D\partial_x^2 = -Dk_x^2$

For these simulations:

$\lambda=0$

```
<output format="ascii" filename="RPS_spde_delT.xsil">

  <sampling_group basis="x y" initial_sample="yes">
    <moments>aR bR cR srtD</moments>
    <dependencies>concentration</dependencies>
    <![CDATA[
        aR = a.Re();
        bR = b.Re();
        cR = c.Re();
      srtD = sqrt((mu*cR*(1-aR-bR-cR) + sig*cR*bR)/N);
    ]]>
  </sampling_group>

 </output>
</simulation>
```